

Weak Identification Robust Methods for Production Function Estimation*

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Abstract

This paper revisits control-function estimation of production functions. We show that, in empirically relevant environments, the structural parameters can be weakly identified even when they are formally point identified. Casting control-function estimators in a GMM framework, we characterize the consequences of weak identification for estimation and inference, including non-standard asymptotic behavior and unreliable Wald inference. We then develop and implement identification-robust inference procedures for control-function estimators and provide practical guidance for diagnosing weak identification and reporting inference that remains valid when identification is weak. Finally, we illustrate how weak identification propagates to economically relevant objects constructed from production-function estimates by studying firm-level markups under the production approach.

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1 Introduction

Production functions are central objects in economics. Their estimation has a long history, yet the core econometric difficulty motivating early work remains central today: firms choose inputs in response to productivity shocks unobserved by the econometrician, rendering inputs endogenous. Control-function methods, developed by Olley and Pakes (1996) (OP), Levinsohn and Petrin (2003) (LP), and Akerberg et al. (2015) (ACF), address this endogeneity by using observable firm decisions—most commonly investment or intermediate inputs—to construct a proxy for latent productivity and deriving moment restrictions from firms’ optimization behavior. These procedures are now used extensively across empirical industrial organization, macroeconomics, labor, and international trade (Akerberg et al., 2007).

Point identification requires the moment conditions to uniquely pin down the structural parameters. Even when this condition is met, however, the moment conditions may contain only limited information about some parameters. Control-function estimators are naturally cast in a GMM framework (Wooldridge, 2009), and the GMM literature—building on Staiger and Stock (1997) and Stock et al. (2002)—has extensively studied what happens when identification is *weak*: the GMM criterion is nearly flat in economically relevant directions, the sampling distribution of the estimator departs substantially from its asymptotic normal approximation, and conventional Wald-type confidence intervals have incorrect coverage (Stock and Wright, 2000; Stock et al., 2002). Formally, weak identification corresponds to the moment Jacobian being near-singular, and identification-robust inference procedures (Stock and Wright, 2000; Kleibergen, 2005) deliver valid inference regardless of identification strength.

In the control-function setting, two distinct sources drive near-singularity of the Jacobian. First, the proxy may carry a weak signal about productivity—a consequence of optimization frictions, heterogeneous input prices, supply-and-demand shocks, or construction from accounting aggregates. Second, near-functional dependence between inputs: because static inputs are freely adjusted in response to productivity, their variation is nearly a deterministic function of dynamic inputs, limiting the independent variation available to separately identify input elasticities. This is the continuous analogue of the exact collinearity that ACF identify as a source of non-identification, and it has the same consequences for inference even when it falls short of exact failure.

This paper makes three contributions. First, we use the moment Jacobian to characterize identification and its strength in the control-function setting. This establishes a precise analogy between

weak proxy informativeness and the weak-instrument problem in linear IV, and provides the basis for the inference procedures developed below. We show that when the Jacobian is nearly rank-deficient, the continuously updated GMM estimator (CUE) has a nonstandard sampling distribution and conventional Wald statistics are size-distorted in ways that standard pre-testing cannot detect. We extend the framework of Stock and Wright (2000) to allow the part of the Jacobian that is due to the strongly identified parameters to depend on the weakly identified parameters, as occurs naturally in the production function model.

Second, we develop an identification-robust inference toolkit that integrates directly into the standard OP/LP/ACF estimation workflow. We implement three complementary procedures: (i) a bootstrap normality diagnostic (Angelini et al., 2024) that tests whether the CUE estimator’s sampling distribution conforms to the asymptotic normal approximation required for conventional inference to be valid; (ii) identification-robust confidence sets obtained by inverting the S-statistic (Stock and Wright, 2000)—the nonlinear GMM analogue of the Anderson-Rubin statistic—which maintain correct coverage regardless of identification strength and apply to the full parameter vector as well as to subsets and linear transformations of interest; and (iii) a weak-identification-robust specification test (Kleibergen, 2005) that remains pivotal under weak identification, enabling us to distinguish limited proxy informativeness from genuine model misspecification—a distinction the standard Hansen J -test cannot make when identification is weak.

Third, we document the empirical relevance of weak identification and show how it propagates to economically important objects. In Monte Carlo experiments calibrated to the ACF firm-dynamics model, the proposed robust procedures maintain correct coverage as proxy informativeness deteriorates, while conventional Wald inference becomes increasingly unreliable. In applications to Chilean plant-level data (ENIA) and U.S. firm-level data (Compustat), identification strength varies systematically with proxy quality: ENIA’s directly measured intermediate consumption yields stable estimates and tight robust confidence sets, whereas Compustat’s accounting-derived materials proxy yields much wider robust sets and strong evidence of nonstandard sampling behavior. We show how this identification uncertainty propagates to production-approach markup estimates, generating economically significant differences between robust and conventional inference that bear on conclusions in the broader markup literature.

The paper proceeds as follows. Section 2 surveys related literature. Section 3 introduces the model. Section 4 characterizes the weak-identification problem. Section 5 develops the inference

toolkit. Section 6 reports Monte Carlo experiments. Section 7.1 presents the empirical applications and markup analysis. Section 8 concludes.

2 Related Literature

We contribute to several strands of literature.

Identification and estimation of production functions. The identification of firm-level production functions from micro data has received sustained attention since Cobb and Douglas (1928), with key contributions by Marschak and Andrews (1944), Blundell and Bond (2000), OP, LP, Wooldridge (2009), ACF, and Gandhi et al. (2020); see Akerberg et al. (2007) for a survey. A central challenge is simultaneity between input choices and unobserved productivity, addressed either by panel-data methods that difference out productivity (Blundell and Bond, 2000) or by control-function methods that proxy for it using observable firm decisions (OP, LP, ACF). Gandhi et al. (2020) show that proxy-based identification can fail when the proxy enters the production function directly, and propose an alternative approach exploiting first-order conditions. Doraszelski and Li (2025) extend the control-function framework to settings with noisy proxies, establishing that identification is retained under an index-sufficiency condition. Our paper takes a different angle on these results. ACF establish that exact functional dependence between inputs renders the production function non-identified; Doraszelski and Li (2025) show that proxy noise need not eliminate identification given index sufficiency. We demonstrate that the phenomena underlying both results remain empirically consequential even when they fall short of causing outright failure: near-functional dependence and limited proxy informativeness lead to weak identification, which generates the same practical difficulties for inference as non-identification.

Weak identification in GMM and structural models. Our contribution is to the literature that documents weak identification in specific structural models. Weak identification has been found in estimation of the New Keynesian Phillips Curve (Mavroeidis, 2005; Nason and Smith, 2008; Dufour et al., 2006; Kleibergen and Mavroeidis, 2009b), monetary-policy rules (Mavroeidis, 2010), dynamic stochastic general-equilibrium (DSGE) models (Ruge-Murcia, 2007; Canova and Sala, 2009; Iskrev, 2010; Andrews and Mikusheva, 2015; Guerron-Quintana et al., 2013), consumption capital asset pricing models (Stock and Wright, 2000), and Euler equations (Yogo, 2004). We add to

this list by documenting weak identification in the control-function production-function framework, characterizing its sources, and showing how the available robust inference tools can be adapted to that setting.

Empirical applications. Control-function estimates of production functions underpin an extensive empirical literature spanning industrial organization, labor, trade, and macroeconomics. They are used to study markups and market concentration (Autor et al., 2020; De Loecker et al., 2020), employer market power (Yeh et al., 2022; Dobbelaere et al., 2024), and firm dynamics, competition, and misallocation (Dobbelaere and Wiersma, 2020; Dobbelaere and Kiyota, 2018; Caselli et al., 2021; Mertens, 2019; Gopinath et al., 2017). A prominent line of research derives firm-level markups from control-function estimates of output elasticities following the wedge logic of Hall (1988) and the firm-level implementation of De Loecker and Warzynski (2012a), generating a large literature on market power and productivity (e.g., De Loecker et al., 2020; Karabarbounis and Neiman, 2018; Traina, 2018; Bridgman and Herrendorf, 2022; Loecker and Scott, 2017; Autor et al., 2020; Melitz and Redding, 2014; Syverson, 2019). Concerns about the reliability of these markup estimates have been raised by Raval (2023), who documents systematic discrepancies between labor- and materials-based markups across datasets, and by De Ridder et al. (2022), who show that with financial-statement data, revenue-based estimates need not identify the level of markups, though they argue that dispersion and time trends remain identified. Our paper raises a distinct concern: weak identification of the underlying production function parameters propagates to markup estimates regardless of data source, affecting inference even on the dispersion and trend objects that De Ridder et al. (2022) consider more robust. We apply our framework to the datasets of Raval (2023) and document how identification uncertainty bears on conclusions drawn for these empirically relevant quantities.

3 Model

We follow the canonical OP/LP/ACF model and cast it in a GMM framework to analyze identification formally.

For firm i in period t , log output satisfies

$$y_{it} = f(k_{it}, v_{it}) + \omega_{it} + \varepsilon_{it}, \tag{1}$$

where k_{it} denotes the d_k predetermined inputs chosen at $t-1$ (e.g. capital), v_{it} denotes d_v static inputs chosen at t (e.g. labor), ω_{it} is the (scalar) firm-specific productivity observed by the firm but not by the econometrician, and ε_{it} is an idiosyncratic shock realized after input choices. This nests Cobb–Douglas, translog, and CES.

Productivity follows a first-order Markov process, such that its conditional distribution depends only on the previous realization.

Assumption 1 (First Order Markov).

$$P(\omega_{it}|\omega_{it-1}, \omega_{it-2}, \dots) = P(\omega_{it}|\omega_{it-1}). \quad (2)$$

This implies that the law of motion can be decomposed into its conditional expectation and an unpredictable innovation:

$$\omega_{it} = g(\omega_{it-1}) + \xi_{it} \quad (3)$$

where $g(\omega_{it}) = \mathbb{E}[\omega_{it}|\omega_{it-1}]$ and the innovation satisfies $\mathbb{E}[\xi_{it}|\omega_{it-1}] = 0$.

Let z_{it} be firm i 's choice vector at t , which includes k_{it+1} and v_{it} . Firms observe ω_{it} when choosing z_{it} ; ε_{it} is realized after these choices and ξ_{it} between $t-1$ and t .

Assumption 2 (Timing and sequential exogeneity). For each firm i , period t , and $\tau \leq t$:

$$\mathbb{E}[\varepsilon_{it} \mid z_{i\tau}] = 0, \quad (4)$$

$$\mathbb{E}[\xi_{it} \mid z_{i\tau-1}] = 0. \quad (5)$$

The key assumption of the control function approach is that observable inputs chosen by the firm serve as proxies for ω_{it} . Let $p_{it} \subset z_{i\tau}$, $\tau \leq t$, denote the proxy vector.

Assumption 3 (Proxy relationship). There exists a function h_t such that

$$\omega_{it} = h_t(p_{it}). \quad (6)$$

This is the scalar-unobservable condition standard in OP/LP/ACF: ω_{it} is fully determined by p_{it} .¹

¹We discuss in Section 4.5 how this condition can be relaxed.

Identification follows two stages. Assumptions 2 and 3 imply the first-stage regression

$$\mathbb{E}[y_{it}|p_{it}] = f(k_{it}, v_{it}) + h_t(p_{it}). \quad (7)$$

Substituting Assumption 3 into the production function gives $y_{it} = f(k_{it}, v_{it}) + h_t(p_{it}) + \varepsilon_{it}$. Taking expectations conditional on p_{it} and applying $\mathbb{E}[\varepsilon_{it} | p_{it}] = 0$ —which follows from Assumption 2 since $p_{it} \subset z_{i\tau}$ for $\tau \leq t$ —yields (7). It is usually imposed that $(k_{it}, v_{it}) \subset p_{it}$, hence f and h_t are not separately identified in this first stage; only the composite is recovered. The second stage uses the Markov law of motion for productivity together with sequential exogeneity to obtain

$$\mathbb{E}[y_{it} | z_{it-1}] = \mathbb{E}[f(k_{it}, v_{it}) + g(\mathbb{E}[y_{it-1}|p_{it-1}] - f(k_{it-1}, v_{it-1})) | z_{it-1}]. \quad (8)$$

Substituting the Markov law of motion (3) into the production function yields $y_{it} = f(k_{it}, v_{it}) + g(\omega_{it-1}) + \xi_{it} + \varepsilon_{it}$. Under Assumption 3, $\omega_{it-1} = h_{t-1}(p_{it-1})$ is z_{it-1} -measurable, and the first stage identifies $h_{t-1}(p_{it-1}) = \mathbb{E}[y_{it-1} | p_{it-1}] - f(k_{it-1}, v_{it-1})$. Taking expectations conditional on z_{it-1} and applying $\mathbb{E}[\xi_{it} | z_{it-1}] = \mathbb{E}[\varepsilon_{it} | z_{it-1}] = 0$ from Assumption 2 yields (8), which provides the basis for identifying f .

3.1 Parametric specification

We parametrize f , g , and h_t using polynomial sieves following Wooldridge (2009). The production function is

$$f(k_{it}, v_{it}) = \alpha_f + x'_{it}\beta,$$

with x_{it} collecting all monomials in (k_{it}, v_{it}) up to degree L_f ; $L_f = 1$ gives Cobb–Douglas and $L_f = 2$ gives translog. The dimension of β is then $d_\beta = \binom{L_f + d_k + d_v}{d_k + d_v} - 1$, since α_f captures the constant separately. The law of motion is

$$g(u) = \alpha_g + b(u)'\rho,$$

with $b(u) = (u, \dots, u^{L_g})'$, so that $\dim(\rho) = L_g$.

The control function is

$$h_t(p_{it}) = \alpha_h + r'_{it}\lambda,$$

where r_{it} collects all monomials in the components of p_{it} up to degree L_h . The dimension of λ is, if $p_{it} \in \mathbb{R}^{d_p}$, $d_\lambda = \binom{L_h + d_p}{d_p} - 1$, again excluding the constant.

Substituting these parametric forms into (7) and (8) yields the estimating equations from which we define the residuals. As mentioned above, the first-stage regression does not identify (β, λ) separately, since x_{it} is a subvector of r_{it} . Instead it identifies the composite parameter $\gamma \in \mathbb{R}^{d_\lambda}$, which satisfies $r'_{it}\gamma = x'_{it}\beta + r'_{it}\lambda$:

$$y_{it} = \alpha_1 + r'_{it}\gamma + u_{1,it}. \quad (9)$$

α_1 is a composite intercept, since the intercepts α_f, α_h are not identified separately. The second-stage equation is

$$y_{it} = \alpha_2 + x'_{it}\beta + b(r'_{it-1}\gamma - x'_{it-1}\beta)\rho + u_{2,it}, \quad (10)$$

where $r'_{it-1}\gamma - x'_{it-1}\beta$ is the observable dynamic index (lagged predicted output net of the production component). Again, α_2 is a composite intercept.

Let $\theta := (\alpha_1, \alpha_2, \beta', \gamma', \rho')' \in \Theta \subset \mathbb{R}^{d_\theta}$ and use (9), (10) to define the stacked residual function $u_{it}(\theta) = (u_{1,it}(\theta), u_{2,it}(\theta))'$. With instrument vectors $w_{1,it} = (1, r'_{it})'$ and $w_{2,it} = (1, \tilde{r}'_{it-1})'$ stacked block-diagonally as W_{it} , the moment contribution is $\phi_{it}(\theta) = W'_{it}u_{it}(\theta)$, where $\tilde{r}_{it-1}$ collects all monomials in r_{it-1} up to degree d_ρ .

3.2 Special case: AR(1)

In the special case of a linear law of motion with $d_\rho = 1$ (AR(1) law of motion), the two equations collapse to the single estimating equation

$$y_{it} = \alpha_s + x'_{it}\beta + \rho(y_{it-1} - x'_{it-1}\beta) + u_{s,it}, \quad (11)$$

which obviates separating identification in two stages separately and reduces the parameter space considerably, by d_λ . We can use (11) to define residual function $u_{s,it}(\theta)$. With instruments $w_{s,it} = (1, r_{it-1})$, the moment contributions are then $\phi_{it}(\theta) = w_{s,it}u_{s,it}(\theta)$.

3.3 GMM Estimation

We estimate the model by GMM. Let $\phi_i(\theta) := \frac{1}{T-1} \sum_{t=2}^T \phi_{it}(\theta)$ aggregate moment contributions to the firm level, and let $\bar{\phi}_n(\theta) := \frac{1}{n} \sum_{i=1}^n \phi_i(\theta)$ be the sample mean. The GMM criterion is

$$Q_n(\theta) := n \bar{\phi}_n(\theta)' W_n \bar{\phi}_n(\theta),$$

where W_n is a positive-definite weighting matrix. We implement either two-step GMM, with $W_n = \hat{V}_{\phi\phi}(\hat{\theta}_1)^{-1}$ where $\hat{\theta}_1$ is a preliminary consistent estimator, or the continuously updated estimator (CUE), with $W_n = \hat{V}_{\phi\phi}(\theta)^{-1}$. The cluster-robust variance estimator,

$$\hat{V}_{\phi\phi}(\theta) := \frac{1}{n} \sum_{i=1}^n (\phi_i(\theta) - \bar{\phi}_n(\theta)) (\phi_i(\theta) - \bar{\phi}_n(\theta))',$$

is consistent for $n \rightarrow \infty$ with T fixed and accommodates arbitrary within-firm serial dependence.

4 Identification Analysis

We use the moment Jacobian to characterize identification of the structural parameters (β, ρ) , establish when identification is weak and what it implies for estimation and inference, and show that formal identification survives nuisance-function misspecification of the proxy relationship.

4.1 Concept of Identification

The model is identified if there exists a unique $\theta_0 \in \Theta$ satisfying the moment conditions

$$\mathbb{E}[\phi_{it}(\theta_0)] = 0.$$

Global identification requires this θ_0 to be the unique solution. Local identification requires that no other parameter in a neighborhood of θ_0 satisfies the conditions, which holds if and only if the population Jacobian

$$\mathcal{J} := \left. \frac{\partial}{\partial \theta'} \mathbb{E}[\phi_{it}(\theta)] \right|_{\theta=\theta_0}$$

has full column rank d_θ (Hansen, 1982; Newey and McFadden, 1994).²

²Local identification governs the curvature of the criterion surface and the precision of estimation. Global identification additionally requires uniqueness of the zero of $\theta \mapsto \mathbb{E}[\phi_{it}(\theta)]$ over Θ ; we maintain this as a high-level assumption

4.2 Identification Failure

When $\mathcal{J}$ is rank-deficient, the moment map is locally flat in at least one direction and distinct parameter values generate identical moment conditions: the structural parameters are not identified.

We characterize the structural sources of this failure in the control-function model.

Jacobian structure in the control-function model. We focus on $\vartheta := (\beta', \rho')'$, since γ is generically identified from the first-stage linear regression.³ The second-stage moment vector is $m(\vartheta) := \mathbb{E}[w_{2,it} u_{2,it}(\vartheta)]$. Using the parametric second-stage equation (10), the gradient of $u_{2,it}$ with respect to ϑ is

$$q_{it}(\vartheta_0) := - \left. \frac{\partial u_{2,it}(\vartheta)}{\partial \vartheta} \right|_{\vartheta_0} = \begin{pmatrix} x_{it} - \dot{b}(r'_{it-1} \lambda_0)' \rho_0 \cdot x_{it-1} \\ b(r'_{it-1} \lambda_0) \end{pmatrix}, \quad (12)$$

where $\dot{b}(u) := \partial b(u) / \partial u = (1, 2u, \dots, L_g u^{L_g-1})'$ and $r'_{it-1} \lambda_0 = h_{t-1}(p_{it-1})$ at true values by definition of λ_0 . The second-stage Jacobian is then

$$\mathcal{J}_2 = \mathbb{E}[w_{2,it} q_{it}(\vartheta_0)']. \quad (13)$$

Sources of rank deficiency. We identify two structural sources of rank deficiency in $\mathcal{J}_2$. We motivate these here.

(i) *Functional dependence.* The β -columns of $\mathcal{J}_2$ are

$$\mathcal{J}_{\beta_j} = \mathbb{E}[w_{2,it} (x_{j,it} - \dot{b}(r'_{it-1} \lambda_0)' \rho_0 x_{j,it-1})].$$

When static inputs obey a policy rule $v_{it} = v(k_{it}, \omega_{it})$,⁴ the quasi-differenced static input is driven by the same state variables as the capital term. The productivity law of motion decomposes variation in v_{it} into a persistent component through ω_{it-1} , which is correlated with predetermined inputs, and the innovation ξ_{it} . Since Assumption 2 implies $\mathbb{E}[\xi_{it} \mid w_{2,it}] = 0$, the innovation contributes nothing to $\mathcal{J}_{\beta_v}$. As a result, $\mathcal{J}_{\beta_v}$ and $\mathcal{J}_{\beta_k}$ may become linearly dependent, or nearly so.

and focus on local identification strength throughout. See Newey and McFadden (1994) for primitive conditions.

³The first stage (9) regresses y_{it} on r_{it} , which consists entirely of predetermined functions of the proxy; all regressors are exogenous by Assumption 2, so identification is generally nonproblematic.

⁴ACF identify functional dependence as the source of non-identification in the first stage of the two-step procedure: because v_{it} is a function of (k_{it}, ω_{it}) and ω_{it} is recovered from the proxy, the first-stage regressors are perfectly collinear and f and h_t cannot be separately identified. The channel here operates at the second stage: even after the two-step fix, functional dependence can create multicollinearity in the Jacobian.

(ii) *Weak proxy.* The ρ -columns of $\mathcal{J}_2$ are

$$\mathcal{J}_\rho = \mathbb{E}[w_{2,it} b(r'_{it-1} \lambda_0)'],$$

where $r'_{it-1} \lambda_0$ is itself a function of lagged inputs. When $b(r'_{it-1} \lambda_0)$ lies in the column space of $\mathcal{J}_\beta$ —because the proxy signal is entirely spanned by the variation used to identify β —the ρ -columns add no independent direction and $\mathcal{J}_2$ is rank-deficient.

Example: Cobb–Douglas and AR(1) law of motion. We use a tractable, fully specified model to verify that the two sources identified above lead to rank failure and to trace the failure to structural primitives. Output follows Cobb–Douglas,

$$y_{it} = \beta_k k_{it} + \beta_\ell \ell_{it} + \omega_{it} + \varepsilon_{it}, \quad (14)$$

with productivity AR(1),

$$\omega_{it} = \rho \omega_{it-1} + \xi_{it}. \quad (15)$$

The proxy is intermediate input m_{it} , with informativeness summarized by the linear conditional mean

$$\mathbb{E}[\omega_{it} \mid k_{it}, m_{it}] = \delta_k k_{it} + \delta_m m_{it}, \quad (16)$$

where δ_m indexes the marginal signal in m_{it} about ω_{it} . The parameter δ_m is the key structural primitive governing identification strength through the weak-proxy channel. Combining (14)–(15) gives the one-step equation (11) with $\theta = (\beta_k, \beta_\ell, \rho)'$ and instrument vector $w_{s,it-1} = (1, k_{it}, k_{it-1}, \ell_{it-1}, m_{it-1})'$. The Jacobian columns are

$$\begin{aligned} \mathcal{J}_{\beta_k} &= \mathbb{E}[w_{s,it-1} (k_{it} - \rho_0 k_{it-1})], \\ \mathcal{J}_{\beta_\ell} &= \mathbb{E}[w_{s,it-1} (\ell_{it} - \rho_0 \ell_{it-1})], \\ \mathcal{J}_\rho &= \mathbb{E}[w_{s,it-1} (y_{it-1} - \beta_{k0} k_{it-1} - \beta_{\ell 0} \ell_{it-1})] = \mathbb{E}[w_{s,it-1} \omega_{it-1}], \end{aligned}$$

where the last equality uses mean-independence of ε_{it-1} .

Functional dependence. Under Cobb–Douglas production function, labor satisfies a linear policy

function in capital and productivity⁵:

$$\ell_{it} = c_0 + c_1 k_{it} + c_2 \omega_{it}. \quad (17)$$

Under AR(1) productivity, $\ell_{it} - \rho_0 \ell_{it-1} = c_1(k_{it} - \rho_0 k_{it-1}) + c_2 \xi_{it}$. Since $\mathbb{E}[\xi_{it} | w_{s,it-1}] = 0$, by Assumption 2, $\mathcal{J}_{\beta_\ell} = c_1 \mathcal{J}_{\beta_k}$, so two columns are exactly collinear: $\text{rank}(\mathcal{J}) \leq 2$.

Weak proxy. Suppose *sluggish capital* ($k_{it} = k_{it-1}$)⁶ and *zero proxy signal* ($\delta_m = 0$). Then $\mathbb{E}[\omega_{it-1} | w_{s,it-1}] = \delta_k \mathbb{E}[k_{it-1} | w_{s,it-1}]$ and $\mathbb{E}[k_{it} - \rho_0 k_{it-1} | w_{s,it-1}] = (1 - \rho_0) \mathbb{E}[k_{it-1} | w_{s,it-1}]$, so $\mathcal{J}_\rho = \frac{\delta_k}{1-\rho_0} \mathcal{J}_{\beta_k}$ and $\text{rank}(\mathcal{J}) \leq 2$.

These conditions—exact functional dependence and a fully uninformative proxy—are knife-edges unlikely to hold literally. They are not, however, as artificial as they may appear: the structural forces that generate them are present in many production settings, and when they hold approximately rather than exactly, weak identification problems arise. Section 4.3 makes this precise.

4.3 Weak Identification

When $\mathcal{J}_2$ is near-singular, the GMM criterion is nearly flat in some directions: the estimator has a nonstandard sampling distribution and conventional Wald inference is unreliable. These distortions arise continuously as identification weakens, well away from the knife-edge, making weak identification a concern in empirically relevant settings.

Sources of near-rank deficiency. The two knife-edge conditions identified in Section 4.2 need not hold exactly to generate these problems; near-violations suffice. We call these *almost functional dependence* and a *weak proxy*.

Almost functional dependence arises when the static input policy rule $v_{it} = v(k_{it}, \omega_{it})$ holds only approximately—for instance because of optimization frictions or measurement error in inputs. The quasi-differenced static input is then nearly, but not exactly, proportional to the corresponding capital term, so columns in $\mathcal{J}_\beta$ are nearly collinear and $\mathcal{J}_2$ is nearly rank-deficient.

A *weak proxy* arises when the proxy variable carries limited information about productivity, so that the proxy is nearly spanned by the variation used to identify β . This can occur because the proxy signal is structurally weak relative to the noise, for instance if it is contaminated by measurement

⁵See Appendix B.1.

⁶LP's findings suggest that capital investments is often for a wide range of industries, lending empirical support to this assumption.

error, which attenuates the signal toward zero. Measurement error in the proxy violates the scalar-unobservable condition of Assumption 3; however, as Section 4.5 shows, this need not preclude identification of f —under a weaker conditional mean condition, identification can survive even when the proxy is noisy.

Example: Cobb–Douglas and AR(1) law of motion (continued). We now quantify the two sources in the Cobb–Douglas/AR(1) setting. We first look at *almost functional dependence*. Following the Monte Carlo design of Section 6, introduce optimization noise into the labor policy:

$$\ell_{it} = \ell_{it}^* + \xi_{it}^l, \quad \xi_{it}^l \sim \mathcal{N}(0, \sigma_{\xi_l}^2),$$

where ℓ_{it}^* is optimal labor demand given by (17) and $\mathbb{E}[\xi_{it}^l \mid z_{i,t-1}] = 0$. Similar steps as in the knife-edge case yield

$$\mathcal{J}_{\beta_\ell} = c_1 \mathcal{J}_{\beta_k} - \rho_0 \mathbb{E}[w_{s,it-1} \xi_{it-1}^l],$$

where $\mathbb{E}[w_{s,it-1} \xi_{it-1}^l] = \sigma_{\xi_l}^2 \mathbf{e}_\ell \neq 0$, since $\ell_{it-1} = \ell_{it-1}^* + \xi_{it-1}^l$ implies $\mathbb{E}[\ell_{it-1} \xi_{it-1}^l] = \sigma_{\xi_l}^2$, while all other components of $w_{s,it-1}$ are uncorrelated with ξ_{it-1}^l , and $\mathbf{e}_\ell$ selects the ℓ_{it-1} coordinate. At $\sigma_{\xi_l} = 0$, ξ_{it}^l drops out and rank failure obtains. As σ_{ξ_l} grows, the independent component of $\mathcal{J}_{\beta_\ell}$ grows and identification improves.

We then turn to the *weak proxy* source. Let the proxy be measured with classical error:

$$m_{it} = m_{it}^* + \nu_{it}, \quad \nu_{it} \sim \mathcal{N}(0, \sigma_\nu^2),$$

where m_{it}^* is true intermediate demand and ν_{it} is mean-independent of $z_{i,t-1}$. Measurement error attenuates the proxy signal in (16): the conditional mean of productivity given observed inputs becomes

$$\mathbb{E}[\omega_{it-1} \mid k_{it-1}, m_{it-1}] = \delta_k k_{it-1} + \tilde{\delta}_m(\sigma_\nu) m_{it-1},$$

where $\tilde{\delta}_m(\sigma_\nu) := \delta_m \cdot \sigma_{m^*}^2 / (\sigma_{m^*}^2 + \sigma_\nu^2)$ and $\sigma_{m^*}^2 = \text{Var}(m_{it}^*)$.⁷ Since $m_{it-1} \in w_{s,it-1}$, we have $\mathbb{E}[w_{s,it-1} m_{it-1}] \neq 0$. Under sluggish capital:

$$\mathcal{J}_\rho = \frac{\delta_k}{1 - \rho_0} \mathcal{J}_{\beta_k} + \tilde{\delta}_m(\sigma_\nu) \mathbb{E}[w_{s,it-1} m_{it-1}].$$

⁷The expression for $\tilde{\delta}_m(\sigma_\nu)$ is the standard attenuation bias formula: classical measurement error shrinks the signal-to-noise ratio $\sigma_{m^*}^2 / (\sigma_{m^*}^2 + \sigma_\nu^2)$ below one, attenuating the proxy coefficient toward zero.

At $\sigma_\nu = 0$, $\tilde{\delta}_m = \delta_m \neq 0$ and $\mathcal{J}_\rho$ has an independent direction. As $\sigma_\nu \rightarrow \infty$, $\tilde{\delta}_m \rightarrow 0$, this direction vanishes and rank failure obtains.

Proxy weakness from measurement error violates the scalar-unobservable condition of Assumption 3, but need not preclude identification of f : Section 4.5 shows that for finite σ_ν , identification can survive under a weaker conditional mean condition even as the proxy becomes noisy.

4.4 Asymptotic theory

When $\mathcal{J}_2$ is nearly rank-deficient, the GMM criterion is nearly flat along weakly identified directions, producing a nonstandard sampling distribution for the estimator and invalidating standard inference procedures. We now make this precise; proofs are in Appendix A.

Let $\vartheta = (\vartheta'_w, \vartheta'_s)' \in \Theta_w \times \Theta_s$, where $\vartheta_w \in \mathbb{R}^{d_w}$ collects the weakly identified directions and $\vartheta_s \in \mathbb{R}^{d_s}$ the strongly identified directions, with $d_w + d_s = d_\beta + d_\rho$. Any subset of the structural parameters may belong to either block. We show that the asymptotic behaviour of the entire parameter ϑ is affected even if the Jacobian is only partially near-rank reduced.

Define the population moments $m_n(\vartheta_w, \vartheta_s) := \mathbb{E}[\bar{\phi}_n(\vartheta_w, \vartheta_s)]$. Using a first-order expansion around $\vartheta_0 = (\vartheta'_{w0}, \vartheta'_{s0})'$,

$$m_n(\vartheta_w, \vartheta_s) = m_n(\vartheta_{w0}, \vartheta_{s0}) + M_{w,n}(\vartheta_w, \vartheta_s)(\vartheta_w - \vartheta_{w0}) + M_{s,n}(\vartheta_w, \vartheta_s)(\vartheta_s - \vartheta_{s0}) + r_n(\vartheta_w, \vartheta_s), \quad (18)$$

where

$$M_{w,n}(\vartheta_w, \vartheta_s) := \mathbb{E}\left[\frac{\partial \bar{\phi}_n(\vartheta_w, \vartheta_s)}{\partial \vartheta'_w}\right], \quad M_{s,n}(\vartheta_w, \vartheta_s) := \mathbb{E}\left[\frac{\partial \bar{\phi}_n(\vartheta_w, \vartheta_s)}{\partial \vartheta'_s}\right].$$

We model a partially nearly rank deficient Jacobian following Stock and Wright (2000) as a sequence of DGPs along which $M_{w,n} = O(n^{-1/2})$ while $M_{s,n}$ remains of fixed order.

Assumption 4 (Partial local-to-zero Jacobian). Along a sequence of DGPs indexed by n ,

- (i) $M_{w,n}(\vartheta_w, \vartheta_s) = n^{-1/2}C(\vartheta_w, \vartheta_s)$, where $C(\vartheta_w, \vartheta_s) \in \mathbb{R}^{d_\phi \times d_w}$ is continuous in ϑ_s and bounded on $\Theta_w \times \Theta_s$.
- (ii) $M_{s,n}(\vartheta_w, \vartheta_s) = R(\vartheta_w)$, where $R(\vartheta_w) \in \mathbb{R}^{d_\phi \times d_s}$ is continuous and bounded on Θ_w .
- (iii) $[C(\vartheta_w, \vartheta_s) \ R(\vartheta_w)]$ has full column rank uniformly on $\Theta_w \times \Theta_s$.
- (iv) $\sup_{(\vartheta_w, \vartheta_s) \in \Theta_w \times \Theta_s} \|r_n(\vartheta_w, \vartheta_s)\| = o(n^{-1/2})$.

Condition (i) formalizes the near rank-reduced Jacobian that vanishes at rate $n^{-1/2}$ in the weakly identified directions. Condition (ii) requires that the strongly identified directions retain an $O(1)$ Jacobian, so those parameters remain well-identified along the sequence. Condition (iii) ensures the problem is *partially* weakly identified: the Jacobian has full column rank, so the model is not fully unidentified. Condition (iv) requires the remainder in the first-order expansion to be $o(n^{-1/2})$, so the linear approximation is exact at the scale of the weak Jacobian.

Assumption 4 extends Stock and Wright (2000) in one respect: the strong-block Jacobian $R(\vartheta_w)$ is allowed to depend on the weakly identified parameters. Stock and Wright (2000) require the strong-block moment function to be independent of the weak block, so their analogue of R is a fixed matrix evaluated at the true parameter value. In the production function model, however, $R(\vartheta_w) = \mathbb{E}[w_{2,it} b(r'_{it-1}\gamma - x'_{it-1}\beta)']$ depends on $\beta = \vartheta_w$, so that assumption is inapplicable. The consequence is that consistency of $\hat{\vartheta}_s$ cannot be established via a fixed population criterion: the separation argument in Proposition 7 must hold uniformly over Θ_w , which requires the uniform full-column-rank condition on $R(\vartheta_w)$ in condition (iii).

Although the estimator might have non-standard distribution, the sample moments on which it is based are asymptotically uniformly Gaussian under weak conditions. Assumption 5 requires a functional CLT for the empirical process; it is implied by standard primitive conditions such as Lipschitz moment functions with square-integrable Lipschitz constants (Andrews, 1994).

Assumption 5 (Uniform Gaussian limit). The centered empirical process $\psi_n(\vartheta) := \sqrt{n}\{\bar{\phi}_n(\vartheta) - m_n(\vartheta)\}$ converges weakly on $\Theta_w \times \Theta_s$ to a mean-zero Gaussian process ψ_0^f with almost surely bounded sample paths.

Assumption 6 is a standard regularity condition on the weighting matrix.

Assumption 6 (Positive definite weighting matrix). $W_n \xrightarrow{p} W_0$ with W_0 symmetric positive definite.

Proposition 7 ($\sqrt{n}$ -consistency of the strongly identified block). *Under Assumptions 4–6, for any GMM minimizer $\hat{\vartheta} = (\hat{\vartheta}'_w, \hat{\vartheta}'_s)'$ of $Q_n(\vartheta) = n \bar{\phi}_n(\vartheta)' W_n \bar{\phi}_n(\vartheta)$:*

$$\hat{\vartheta}_s \xrightarrow{p} \vartheta_{s0} \quad \text{and} \quad \sqrt{n}(\hat{\vartheta}_s - \vartheta_{s0}) = O_p(1).$$

Since Proposition 7 establishes $\sqrt{n}$ -consistency of $\hat{\vartheta}_s$, the behavior of the criterion along weakly identified directions is governed by $\sqrt{n}$ -local neighborhoods of ϑ_{s0} . It therefore suffices to study Q_n at perturbations of the form $\tilde{\vartheta}_s = \vartheta_{s0} + \lambda/\sqrt{n}$ for $\lambda \in \mathbb{R}^{d_s}$, so that $\sqrt{n}(\tilde{\vartheta}_s - \vartheta_{s0}) = O(1)$.

Proposition 8 (Uniform boundedness of the criterion). *Under Assumptions 4–6:*

$$\sup_{\vartheta_w \in \Theta_w} Q_n(\vartheta_w, \tilde{\vartheta}_s) = O_p(1).$$

Proposition 8 is the analogue of Stock and Wright (2000, Theorem 1) for the partially weakly identified GMM setting. Because $Q_n(\vartheta_w, \tilde{\vartheta}_s)$ is stochastically bounded uniformly over Θ_w , the weakly identified component $\hat{\vartheta}_w = O_p(1)$ has a generally non-normal limiting distribution. The strongly identified block remains consistent, but its limiting distribution is also generally non-normal because it depends on the random limit of $\hat{\vartheta}_w$. Conventional Wald inference is therefore unreliable, which motivates the identification-robust procedures in Section 5.

4.5 Nuisance-Function Misspecification

Assumption 3 imposes exact invertibility: $\omega_{it} = h_t(p_{it})$. In practice, measurement error, optimization noise, or input-price heterogeneity cause the proxy to recover productivity only partially. This misspecifies the canonical nuisance structure, but need not preclude identification of f .

Replace Assumption 3 with the conditional mean condition

$$\mathbb{E}[\omega_{it} \mid p_{it}] = h_t(p_{it}). \tag{19}$$

This defines the proxy error $\eta_{it} := \omega_{it} - h_t(p_{it})$ which satisfies $\mathbb{E}[\eta_{it} \mid p_{it}] = 0$. Under exact invertibility, the canonical law of motion gives $\mathbb{E}[\omega_{it} \mid p_{it-1}] = g(h_{t-1}(p_{it-1}))$. Under (19), ω_{it-1} is no longer exactly recovered from p_{it-1} , so this equality fails and g ceases to be directly identifiable. Doraszelski and Li (2025) show that identification of f is nonetheless preserved if the conditional mean of productivity given the proxy index satisfies an index-sufficiency condition.

Assumption 9 (Index sufficiency). There exists a link function $\tilde{g}(\cdot)$ such that $\mathbb{E}[\omega_{it} \mid p_{it-1}] = \tilde{g}(h_{t-1}(p_{it-1}))$.

$\tilde{g}$ is a reduced-form object: it characterizes the conditional mean of productivity given the proxy index, rather than the structural transition g . Substituting Assumption 9 into the production function gives

$$y_{it} = f(k_{it}, v_{it}) + \tilde{g}(h_{t-1}(p_{it-1})) + u_{2,it}^*,$$

where $u_{2,it}^* := \eta_{it-1} + \xi_{it} + \varepsilon_{it}$ and $\eta_{it-1} := g(\omega_{it-1}) - \tilde{g}(h_{t-1}(p_{it-1}))$ satisfies $\mathbb{E}[\eta_{it-1} \mid p_{it-1}] = 0$

by Assumption 9. Under nuisance-function misspecification, the canonical structural decomposition fails, but the second-stage moment conditions $\mathbb{E}[w_{2,it} u_{2,it}^*] = 0$ still hold for the modified composite error, so f remains identified.

The weak-proxy channel of Section 4.3 is therefore best understood as degradation of identification strength rather than identification failure: as proxy noise grows, the Jacobian weakens continuously, producing the near-rank deficiency analyzed above, while formal identification is preserved.

5 Identification Robust Methods

We employ identification-robust inference procedures for the production function parameters within the control-function approach. Conventional inference methods, such as Wald statistics and t -statistics, rely on asymptotic normality, which breaks down under weak identification, resulting in nonstandard distributions and size distortions (Staiger and Stock, 1997; Stock and Wright, 2000). Selecting instruments or proxy variables based on standard pre-testing, such as the commonly used F -test, does not resolve the issue, as it may introduce selection bias resulting in a test that is not size-controlled.

Identification-robust statistics overcome these problems by maintaining a valid limiting distribution regardless of identification strength, ensuring reliable hypothesis testing and confidence intervals (Andrews and Stock, 2005). These statistics are defined for the CUE.

5.1 Rank test for identification

A straightforward approach to diagnosing identification strength is to directly test whether the Jacobian $\mathcal{J}$ has full column rank. The null hypothesis is that the Jacobian has exactly one rank deficiency:

$$H_0 : \text{rank}(\mathcal{J}) = d_\theta - 1,$$

against the alternative that $\text{rank}(\mathcal{J}) = d_\theta$. We test this using the minimum eigenvalue of the sample concentration matrix evaluated at the CUE estimate $\hat{\theta}$:

$$T_{\text{CD}} = \lambda_{\min}\left(n \hat{\mathcal{J}}_n(\hat{\theta})' \hat{V}_{\phi\phi}(\hat{\theta})^{-1} \hat{\mathcal{J}}_n(\hat{\theta})\right), \quad (20)$$

where $\widehat{\mathcal{J}}_n(\hat{\theta}) = \partial \bar{\phi}_n(\theta) / \partial \theta' |_{\hat{\theta}}$. This is the nonlinear GMM analogue of the Cragg–Donald minimum-eigenvalue statistic (Cragg and Donald, 1993). Under H_0 , $T_{\text{CD}} \xrightarrow{d} \chi^2_{d_\phi - d_\theta + 1}$. Large values reject H_0 and signal identification; failure to reject signals potential underidentification. In the just-identified case ($d_\phi = d_\theta$) the degrees of freedom equal one.

In nonlinear GMM, $\widehat{\mathcal{J}}_n(\theta)$ depends on θ and $\hat{\theta}$ is inconsistent under H_0 . Wright (2003) shows that the formally valid procedure eliminates θ by minimizing $T_{\text{CD}}(\theta)$ over the Stock–Wright S-set, yielding a statistic $L^* = \inf_{\theta \in S^*(\alpha)} T_{\text{CD}}(\theta)$ that is asymptotically conservative with size $\leq 2\alpha$. Since $L^* \leq T_{\text{CD}}$ by construction, our diagnostic is *favourable* towards finding identification: whenever T_{CD} rejects H_0 , so does L^* . Failure to reject T_{CD} therefore constitutes strong evidence of underidentification.

Beyond the minimum eigenvalue, the full set of eigenvalues of $n \widehat{\mathcal{J}}_n(\hat{\theta})' \widehat{V}_{\phi\phi}(\hat{\theta})^{-1} \widehat{\mathcal{J}}_n(\hat{\theta})$ — one per parameter — is informative: when identification is strong all eigenvalues diverge, while a near-zero eigenvalue in a particular direction reveals which linear combination of parameters is poorly identified.

5.2 Bootstrap pre-test for proxy relevance

While the rank test assesses identification at the level of the population Jacobian, it does not directly reveal whether the resulting distortion to the estimator’s sampling distribution is empirically consequential. Angelini et al. (2024) propose a bootstrap normality diagnostic that addresses this gap. Rather than examining the Jacobian, it tests whether the CUE’s finite-sample distribution conforms to the asymptotic normal approximation required for conventional inference to be valid—the key advantage being that it is symptom-based and therefore sensitive to the full downstream effect of weak identification, including nonlinearities that a Jacobian rank test alone may not capture. The null hypothesis is that all proxies (and instruments) are strong.

Let $\hat{\theta}_n$ denote the CUE obtained from the full sample and let V_θ be a consistent estimate of its asymptotic covariance, which we can compute from $\widehat{V}_{\phi\phi}(\hat{\theta})$. Let $\hat{\theta}_n^*$ be the bootstrapped counterpart, using the Moving Block Bootstrap (MBB) technique. The MBB is similar in spirit to a standard bootstrap with replacement, but instead of resampling one observation at a time it resamples blocks of observations—in our case at the firm level—in order to replicate their serial dependence structure

(Angelini et al., 2024). We can then form

$$\Gamma_n^* = \sqrt{n} V_\theta^{-1/2} (\hat{\theta}_n^* - \hat{\theta}_n).$$

Under the null, Γ_n^* is asymptotically standard normal. Testing multivariate normality—e.g. with the Doornik–Hansen or a set of Kolmogorov–Smirnov statistics—therefore provides a valid decision rule. Failure to reject normality implies that standard Wald inference remains appropriate; rejection signals weak proxies, in which case one should switch to identification-robust procedures such as the AR-GMM confidence sets introduced below. In light of Proposition 8, weak proxies lead to a nonregular, non-Gaussian limit for $\hat{\theta}_n$; the bootstrap diagnostic exploits this by testing whether the studentized bootstrap increment remains asymptotically Gaussian, as it must under strong identification.

The pre-test does not distort subsequent inference because the bootstrap normality statistic is asymptotically independent of any post-test identification-robust statistic. Under the null of strong proxies, Γ_n^* has a nonrandom asymptotic law conditional on the sample. As a consequence, the diagnostic decision is asymptotically independent of any statistic computed from the original data. Conditioning subsequent inference on nonrejection therefore does not alter its asymptotic rejection probabilities (or coverage).

Moreover, this test is straightforward to implement: aside from the re-estimations needed for the bootstrap, no extra first-stage regressions or critical-value adjustments are required, and the method accommodates multiple proxies, conditional heteroskedasticity, and zero-censored instruments. In practice we recommend reporting both the rank diagnostic and the bootstrap pre-test: the former to diagnose identification strength in the Jacobian, the latter to confirm the downstream effect on the estimator’s distribution.

5.3 AR-GMM test

A straightforward identification-robust statistic is the GMM extension of the AR test (Anderson and Rubin, 1949), the AR-GMM. It is based on the S-statistic of Stock and Wright (2000). Let $\theta \in \mathbb{R}^{d_\theta}$ be the parameter of interest and $\bar{\phi}_n(\theta) \in \mathbb{R}^{d_\phi}$ the sample moments with which it is estimated. To test the null hypothesis $H_0 : \theta_0 = \theta^*$, the S-statistic is equivalent to the CUE objective function

evaluated at θ^* :

$$S(\theta^*) = n \bar{\phi}_n(\theta^*)' \widehat{V}_{\phi\phi}(\theta^*)^{-1} \bar{\phi}_n(\theta^*). \quad (21)$$

The S-statistic converges under H_0 to a $\chi_{d_\phi}^2$, irrespective of identification strength (see Theorem 2 in Stock and Wright, 2000).

The AR-GMM test has power against the alternative $\theta \neq \theta^*$ and against violations of moment restrictions in overidentified models. Its power diminishes as the number of moment restrictions increases, since the degrees of freedom of its limiting distribution equal the number of moment conditions. Consequently, when the number of instruments substantially exceeds the number of structural parameters, the statistic exhibits low power. To address this limitation, alternative, more powerful identification-robust test statistics have been developed (Moreira, 2003; Kleibergen, 2002, 2005). We implement the KLM statistic and the weak-identification-robust J -test of Kleibergen (2005) in our empirical applications; see Sections 5.4 and 5.4.1 below.

5.3.1 Robust Confidence Sets

From Proposition 2 it follows that the $100 \times (1 - \alpha)\%$ confidence set for θ is

$$CS(\alpha) = \{\theta^* : S(\theta^*) \leq \chi_{d_\phi}^2(1 - \alpha)\}, \quad (22)$$

where $\chi_{d_\phi}^2(1 - \alpha)$ denotes the $(1 - \alpha)$ quantile of the $\chi_{d_\phi}^2$ distribution. Since $S(\theta^*)$ is not a quadratic function of θ^* , the confidence set cannot be expressed as an estimator $\pm$ a multiple of a standard error. Instead, one must evaluate $S(\theta^*)$ on a grid of θ^* values and retain those that satisfy the inequality.

The shape of $CS(\alpha)$ is informative about identification. Bounded convex sets indicate strong identification. Bounded but non-convex sets may still signal identification difficulties. Unbounded sets arise under weak identification, when the criterion is nearly flat in some direction. Empty sets indicate that no parameter value satisfies the moment conditions, pointing to model misspecification. Wang and Zivot (1998) discuss these shapes in the bivariate AR case: elliptical sets correspond to strong identification, while hyperbolic sets indicate potential weak identification.

Confidence sets for a subvector of θ can be obtained by substituting the remaining elements with their estimates, then computing $S(\theta^*)$ over a grid varying only the subvector of interest. An

alternative is to project a valid joint confidence set for θ onto the subspace of interest (Dufour, 1997, Sec. 5.2), which yields an asymptotically conservative confidence set.

5.3.2 AR-GMM tests on linear transformations of the parameter vector

In many applications the objects of interest are linear combinations of the structural parameters rather than the parameters themselves. Let $\theta \in \mathbb{R}^{d_\theta}$ denote the original parameter vector and

$$\psi := R\theta \in \mathbb{R}^{d_\psi}, \quad d_\psi \leq d_\theta,$$

be a parameter of interest defined by a known matrix R of full row rank. The remaining components of θ are treated as nuisance. Any one-to-one linear reparameterization of θ leaves the S-statistic and its limiting distribution invariant, so inference on ψ can be carried out by working directly with the original GMM objective.⁸

Write $\theta = (\psi', \eta')'$ after a suitable reparameterization, where $\eta \in \mathbb{R}^{d_\eta}$ with $d_\eta = d_\theta - d_\psi$ collects the nuisance components. To test $H_0 : \psi = \psi^*$, define the *restricted* CUE estimator of η as

$$\hat{\eta}(\psi^*) := \arg \min_{\eta} S(\psi, \eta) \big|_{\psi=\psi^*}.$$

The corresponding linear-transformation S-statistic is $S(\psi^*, \hat{\eta}(\psi^*))$.

Under H_0 , the limiting null distribution of $S(\psi^*)$ is stochastically dominated by a $\chi_{d_\psi-d_\eta}^2$, with equality when η is well identified.⁹ Using the $\chi_{d_\psi-d_\eta}^2(1-\alpha)$ critical value therefore yields an asymptotically size-correct test that remains valid irrespective of the identification strength of η .

An identification-robust $(1-\alpha) \times 100\%$ confidence set for ψ is obtained by inversion:

$$CS_\psi(\alpha) := \left\{ \psi \in \mathbb{R}^{d_\psi} : S(\psi, \hat{\eta}(\psi)) \leq \chi_{d_\psi-d_\eta}^2(1-\alpha) \right\}.$$

In practice, computation proceeds by gridding over ψ , solving the restricted CUE problem for $\hat{\eta}(\psi)$ at each grid point, and retaining all values that satisfy the inequality. As in the full-parameter case, the resulting confidence set may be bounded, unbounded, or disconnected in the presence of weak identification.

⁸See, for example, Kleibergen and Mavroeidis (2009a) for a detailed discussion in linear IV and GMM settings.

⁹Formally, $S(\psi^*) \stackrel{d}{\Rightarrow} Q$, where Q is bounded above in the convex order by $\chi_{d_\psi-d_\eta}^2$ uniformly over parameter sequences that allow for weak identification; see Kleibergen and Mavroeidis (2009a, Thm. 2).

Subset confidence sets are obtained by taking $\psi = \theta_1$ and $\eta = \theta_2$ in the partition $\theta = (\theta'_1, \theta'_2)'$ with $R = (I_{d_{\theta_1}}, 0)$. Subset AR-GMM procedures are (weakly) more powerful than projection-based methods: non-rejection by the linear-transformation test implies non-rejection by the test obtained from projecting the full-parameter AR-GMM confidence set onto the ψ -space, so projection-based sets are typically more conservative (Dufour, 1997).

5.4 Alternative identification-robust tests

The AR-GMM statistic is fully identification-robust but can have limited power in overidentified designs, since its limiting distribution is $\chi^2_{d_\phi}$ regardless of identification strength: more moment conditions mean more degrees of freedom and a higher critical value even when the extra moments contain little information about θ . Kleibergen (2005) proposes a more powerful alternative, the K statistic, which exploits the Jacobian to sharpen inference while retaining a valid limiting distribution under weak identification.

The K statistic. The standard GMM LM statistic of ? uses the sample Jacobian $q_T(\theta_0, Y) = \partial f_T(\theta_0, Y)/\partial \theta'$ as an estimator of the expected Jacobian $J_\theta(\theta_0)$. The key problem is that the limiting behavior of $q_T(\theta_0, Y)$ is *not* independent of the limiting behavior of the moments $f_T(\theta_0, Y)$ when $J_\theta(\theta_0)$ is weak or zero, so the LM statistic's limiting distribution depends on nuisance parameters under weak identification.

Kleibergen (2005) replaces q_T with an estimator $\hat{D}_T(\theta_0, Y)$ constructed from the first-order derivative of the CUE objective function. Specifically, the CUE first-order condition implies

$$\left. \frac{1}{2} \frac{\partial Q(\theta)}{\partial \theta'} \right|_{\theta_0} = f_T(\theta_0, Y)' \hat{V}_{ff}(\theta_0)^{-1} \hat{D}_T(\theta_0, Y),$$

where

$$\hat{D}_T(\theta_0, Y) = \left[q_{1,T}(\theta_0, Y) - \hat{V}_{\theta f,1}(\theta_0) \hat{V}_{ff}(\theta_0)^{-1} f_T(\theta_0, Y), \dots \right],$$

with $\hat{V}_{\theta f,i}(\theta_0)$ estimating the cross-covariance between the i -th column of q_T and f_T . The subtraction of $\hat{V}_{\theta f} \hat{V}_{ff}^{-1} f_T$ orthogonalizes the Jacobian estimator with respect to the moments. By Lemma 1 of Kleibergen (2005), $\hat{D}_T(\theta_0, Y)$ converges in distribution to a limit that is *independent* of the limit of $f_T(\theta_0, Y)$, regardless of whether $J_\theta(\theta_0)$ has full rank, is weak, or is zero. This independence is the key property that gives the K statistic its robustness.

The K statistic for $H_0 : \theta = \theta_0$ is then (Definition 1, Kleibergen, 2005):

$$K(\theta_0) = \frac{1}{T} f_T(\theta_0, Y)' \widehat{V}_{ff}(\theta_0)^{-1/2} P_{\widehat{V}_{ff}(\theta_0)^{-1/2} \widehat{D}_T(\theta_0, Y)} \widehat{V}_{ff}(\theta_0)^{-1/2} f_T(\theta_0, Y), \quad (23)$$

where $P_A = A(A'A)^{-1}A'$ denotes the projection onto the column space of A . Equivalently, writing $u = \sqrt{T} \widehat{V}_{ff}^{-1/2} \bar{f}$ and $A = \widehat{V}_{ff}^{-1/2} \widehat{D}_T$, we have $K(\theta_0) = u' P_A u$. Since $\widehat{D}_T$ is asymptotically independent of f_T , the matrix P_A is asymptotically independent of u , and the projection $u' P_A u$ converges to a $\chi_{d_\theta}^2$ distribution under H_0 irrespective of identification strength (Kleibergen, 2005, Theorem 1). Under strong identification, $K(\theta_0)$ is asymptotically equivalent to the standard GMM LM statistic and therefore has higher power than the S-statistic when there are many overidentifying restrictions.

Subset K for individual parameters and returns to scale. In our application we are often interested in testing a hypothesis on a subvector of θ . Partition $\theta = (\alpha', \beta')'$ where β is the parameter of interest (e.g. a single elasticity or the sum of elasticities for RTS) and α collects the nuisance parameters. Following Kleibergen (2005), Section 3.2, the subset K statistic for $H_0 : \beta = \beta_0$ is obtained by projecting u and the columns of A corresponding to β onto the orthogonal complement of the columns of A corresponding to α :

$$K_\beta(\beta_0) = \tilde{u}' P_{\tilde{A}_\beta} \tilde{u},$$

where $\tilde{u} = M_{A_\alpha} u$, $\tilde{A}_\beta = M_{A_\alpha} A_\beta$, and $M_{A_\alpha} = I - P_{A_\alpha}$ is the annihilator of A_α . This partialling out removes the contribution of the nuisance directions and ensures that K_β converges to $\chi_{d_\beta}^2$ under H_0 and Assumption 3 of Kleibergen (2005) (that the nuisance Jacobian J_α has full rank). Confidence sets for β are obtained by inverting the test over a grid of β_0 values. For RTS, the constraint $\phi = \sum_v \beta_v$ defines a one-dimensional subspace, and we invert the subset K statistic over a grid of ϕ_0 values while profiling out the remaining elasticities.

Spurious power decline and the J statistic. The K statistic has one important limitation. Because K is a quadratic form in the first-order derivative of the CUE objective function $Q(\theta)$, it equals zero wherever $Q(\theta)$ is stationary: at its minimum (the CUE), at any local maximum, and at inflection points. At the CUE the moment conditions are (approximately) satisfied, so $K = 0$ there is correct. At a local maximum or inflection point, however, the moment conditions are

violated—these are values of θ at which the data are inconsistent with the model—yet K still equals zero, falsely failing to reject H_0 at those points. This is the “spurious power decline” of Kleibergen (2005), Section 5: the K confidence set may include regions that are spurious because the moment conditions do not hold there.

The S -statistic $S(\theta_0) = (1/T) Q(\theta_0)$ is zero only at the CUE. Since $S = K + J$ (Kleibergen, 2005, eq. 30), where

$$J(\theta_0) = \frac{1}{T} f_T(\theta_0, Y)' \widehat{V}_{ff}(\theta_0)^{-1/2} M_{\widehat{V}_{ff}(\theta_0)^{-1/2} \widehat{D}_T(\theta_0, Y)} \widehat{V}_{ff}(\theta_0)^{-1/2} f_T(\theta_0, Y), \quad (24)$$

the J statistic captures the part of the criterion not explained by the K direction. Where K is spuriously small—at maxima and inflection points of Q — J is correspondingly large, because S is large at those points. Under H_0 and correct specification, $J(\theta_0) \xrightarrow{d} \chi_{d_\phi - d_\theta}^2$ independently of $K(\theta_0)$ (Kleibergen, 2005, Theorem, p. 1112): K tests the hypothesis while J tests moment validity, and the two are asymptotically independent.

Combined J–K confidence sets. The asymptotic independence of J and K motivates combining them to eliminate the spurious regions from the K confidence set (Kleibergen, 2005, Section 6). For a nominal level α , choose α_K and α_J with $\alpha_K + \alpha_J \approx \alpha$ and define the J–K confidence set as

$$CS_{JK}(\alpha) = \{\theta_0 : K(\theta_0) \leq \chi_{d_\theta}^2(1 - \alpha_K)\} \cap \{\theta_0 : J(\theta_0) \leq \chi_{d_\phi - d_\theta}^2(1 - \alpha_J)\}.$$

The J pretest prunes exactly the spurious values: wherever K is small only because the moment conditions are violated, J is large and the intersection excludes that point. In our empirical applications we use $\alpha_K = 0.04$ and $\alpha_J = 0.01$ for an overall level of $\alpha \approx 0.05$, and we report Subset- JK confidence sets for each individual elasticity and for RTS by applying the subset K inversion described above within this combined framework.

5.4.1 Identification-Robust Specification Test

As established in Section 4.5, noisy proxies do not invalidate the moment conditions: under Assumptions 2–9, identification is retained regardless of proxy error, following Doraszelski and Li (2025). Wide identification-robust confidence sets are therefore consistent with a correctly specified model in which the proxy carries limited information about productivity—they signal weak identification,

not misspecification. To separate these two explanations, we need a specification test that remains valid under weak identification.

The standard Hansen J -statistic,

$$J_H(\hat{\theta}) = S(\hat{\theta}) = n \bar{\phi}_n(\hat{\theta})' \widehat{V}_{\phi\phi}(\hat{\theta})^{-1} \bar{\phi}_n(\hat{\theta}),$$

is $\chi^2_{d_\phi - d_\theta}$ under correct specification and strong identification, but its limiting distribution is distorted when identification is weak because the rate of convergence of $\hat{\theta}$ depends on identification strength (Stock and Wright, 2000; Kleibergen, 2005).

Kleibergen (2005) proposes a J -statistic that remains pivotal under weak identification. Define the standardized Jacobian

$$\tilde{\mathcal{J}}_n(\theta) = \widehat{V}_{\phi\phi}(\theta)^{-1/2} \widehat{\mathcal{J}}_n(\theta), \quad \widehat{\mathcal{J}}_n(\theta) = \frac{\partial \bar{\phi}_n(\theta)}{\partial \theta'},$$

and let

$$M_{\tilde{\mathcal{J}}(\hat{\theta})} = I_{d_\phi} - \tilde{\mathcal{J}}_n(\hat{\theta}) \left(\tilde{\mathcal{J}}_n(\hat{\theta})' \tilde{\mathcal{J}}_n(\hat{\theta}) \right)^{-1} \tilde{\mathcal{J}}_n(\hat{\theta})'$$

be the annihilator of $\tilde{\mathcal{J}}_n(\hat{\theta})$. The *Kleibergen J -statistic* is

$$KJ(\hat{\theta}) = n \bar{\phi}_n(\hat{\theta})' \widehat{V}_{\phi\phi}(\hat{\theta})^{-1/2} M_{\tilde{\mathcal{J}}(\hat{\theta})} \widehat{V}_{\phi\phi}(\hat{\theta})^{-1/2} \bar{\phi}_n(\hat{\theta}). \quad (25)$$

Equivalently, $KJ(\hat{\theta}) = S(\hat{\theta}) - K(\hat{\theta})$, where $K(\hat{\theta})$ is the KLM statistic of Section 5.4 evaluated at $\hat{\theta}$. Under correct specification, $KJ(\hat{\theta}) \xrightarrow{d} \chi^2_{d_\phi - d_\theta}$ irrespective of identification strength (Kleibergen, 2005). Moreover, $KJ(\hat{\theta})$ and $K(\hat{\theta})$ are asymptotically independent under H_0 : $K(\hat{\theta})$ assesses estimation precision (analogous to a first-stage F -statistic) while $KJ(\hat{\theta})$ tests overidentifying restrictions.

The practical interpretation in our setting is as follows. Failure to reject KJ combined with wide AR-GMM confidence sets is consistent with a correctly specified but weakly identified model—the moment conditions are valid, but the proxy provides limited leverage for pinning down the elasticities. Rejection of KJ points to genuine misspecification of the moment conditions irrespective of identification strength. We report KJ alongside the conventional Hansen J in our empirical applications to sharpen this distinction.

6 Monte Carlo Experiment

Our Monte Carlo design follows that of ACF closely. It builds on the analytically solvable dynamic investment model Van Biesebroeck (2003). The parameters are calibrated to reproduce key moments in the Chilean plant data analyzed by Levinsohn and Petrin (2003).

6.1 Setting

Output is produced with a Leontief production function in materials:

$$Y_{it} = \min \left\{ \alpha K_{it}^{\beta_k} L_{it}^{\beta_l} \exp(\omega_{it}), \beta_m M_{it}^* \right\} \exp(\varepsilon_{it}) \quad (26)$$

with $(\alpha, \beta_k, \beta_l, \beta_m) = (1, 0.4, 0.6, 1)$. Y_{it} denotes the output of firm i in period t ; K_{it} , L_{it} and M_{it}^* are, respectively, the levels of capital, labour and material inputs. Note that the material inputs are latent since their observations will be ridden with measurement error. ε_{it} is normal with mean zero with standard deviation 0.01. ω_{it} is assumed to follow an AR(1) with auto-regressive coefficient $\rho = 0.7$.

$$\omega_{it} = \rho \omega_{it-1} + \xi_{it} \quad (27)$$

ε_{it} is independent to the current information set and mean-zero. The variance of the normally distributed innovation ξ_{it} (σ_ξ^2) and the initial value ω_{i0} ($\sigma_{\omega_{i0}}^2$) are set such that the standard deviation of ω_{it} is equal to 0.3.

The capital stock evolves as

$$K_{it} = (1 - \delta) K_{it-1} + I_{it-1}.$$

where I_{it} denotes capital investment, and depreciation rate is set to $\delta = 0.8$. Firms choose investment to maximise the expected, discounted value of future profits subject to convex capital-adjustment costs. Labor and material inputs are selected contemporaneously with output in period t .

Because materials are used in a fixed proportion to output, the “structural value-added” regression equation (see Gandhi et al. (2020)) is

$$y_{it} = \alpha + \beta_k k_{it} + \beta_l l_{it} + \omega_{it} + \varepsilon_{it} \quad (28)$$

where lowercase letters denote the logs. The Leontief specification implies that in optimality we have that both arguments of the min-function have to be equal. This gives us the intermediate demand function.

$$m_{it}^* = -\log(\beta_m) + \alpha + \beta_k k_{it} + \beta_l l_{it} + \omega_{it}, \quad (29)$$

where lower case letters denote the logs. We allow for independent measurement error in m_{it}^* .

$$m_{it} = m_{it}^* + \nu_{it}, \quad (30)$$

where ν_{it} is normal with zero mean and standard deviation σ_ν . We will generate data for different values of σ_ν , which adds noise to the control function. For high values of σ_ν we have weak identification, as will be shown in the results.

As ACF point out, the firm's choice on labor input L_{it} is functionally dependent on, or, in the case of measurement error, strongly correlated with, K_{it} and M_{it} . While this dependence can be accommodated in the second-stage moments, it leaves empirical identification more fragile. To strengthen identification, ACF add independent optimization error to the labor input.

$$l_{it} = l_{it}^* + \xi_{it}^l$$

where ξ_{it}^l follows a normal distribution with mean zero and a standard deviation of σ_{ξ_l} .

6.2 Simulation Results

We consider two DGPs that differ only in the variance of the labor optimization error. The first replicates ACF's DGP with $\sigma_{\xi_l} = 0.37$ (see the description of DGP2 in ACF, Section 5); the second is identical except that the labor optimization error is reduced to $\sigma_{\xi_l} = 0.10$. Effectively this reduces the optimization error in l from around 10% to 5%. This reduction will make the consequences of weak identification more salient at lower measurement error, which serves for illustration. For both DGPs we add classical measurement error in the measured materials input of 0%, 10%, 20%, 50%, where a 10% measurement error raises the variance of observed materials by 10%.

For each simulation, we estimate (β_l, β_k) using the estimation techniques of LP and ACF, and our one-step CUE estimator from Section 7.1. We report the conventional (non-robust) Wald test

and the identification-robust AR-GMM test from Section 5 for the null at the true parameter vector. In addition, we apply the bootstrap normality test of Angelini et al. (2024), also described in from Section 5, which assesses joint normality of the estimators using Shapiro–Wilk–based diagnostics.

Table C.1 reports Monte Carlo estimates under the first DGP ($\sigma_{\xi_t} = 0.37$). With no measurement error, all three estimators (ACF, LP, CUE) are tightly centered at the true values (0.6, 0.4) and exhibit small dispersion. As the error variance increases, LP shows a clear location drift—estimates for β_l move upward while those for β_k move downward—whereas ACF remains comparatively stable, with only a mild upward drift in β_l and negligible movement in β_k . CUE stays closest to unbiased across all error levels, with only moderate increases in variance. Figure D.1 displays the empirical densities of the simulated estimates with normal overlays. It shows that the empirical densities of all estimates are close to normal at 0% error and remain near-normal as the error increases. This is also reflected when testing. Table C.2 shows that rejection frequencies at the 5% level are close to nominal for both the conventional Wald test and the identification-robust AR-GMM test across all error levels; the Doornik–Hansen joint-normality test rejects somewhat more often as measurement error grows, indicating mild departures from normality in this design. It points toward the normality test being conservative.

Table C.3 reports Monte Carlo estimates under the second DGP ($\sigma_{\xi_t} = 0.10$). With no measurement error, all three estimators (ACF, LP, CUE) are centered at the true values and exhibit small dispersion. As the error variance increases to 10%–20%, identification weakens: LP becomes unstable—estimates for β_l move toward one while those for β_k reach implausible magnitudes pointing toward instability—whereas ACF deteriorates more gradually, with β_l drifting up and β_k drifting down and occasional extreme realizations; CUE remains the most stable over this range with a gradual increase in variance. At 50% error, LP exhibits boundary pile-ups and ACF produces explosive draws in a nontrivial share of replications, while CUE continues to yield interpretable coefficients, albeit with thicker tails and higher dispersion. Figure D.2 displays the empirical densities of the simulated estimates. At 0% error the densities are approximately normal and centered at the truth; as measurement error rises, LP’s densities shift (right for β_l , left for β_k) and become spiky near the boundary, ACF develops heavier tails with the same directional drift. While CUE is the most stable, at 20% measurement error and above its sampling distribution shows clear deviations from normality—visible skewness and heavy tails—despite estimates remaining centered. Consistent with these distributional changes, Table C.4 shows that the conventional Wald test increasingly over-rejects

invalidating inference; whereas the identification-robust AR–GMM test maintains rejection frequencies close to 5% across error levels. The Doornik–Hansen joint-normality rejection rate rises sharply with measurement error, consistent with the pronounced departures from normality in this design and indicating that the diagnostic is appropriately sensitive.

7 Empirical Analysis

Our empirical analysis proceeds in two steps. We first estimate production elasticities and implied returns to scale under the ACF baseline and under CUE, using Chilean survey data (ENIA) and U.S. financial-statement data (Compustat). We then combine the estimated elasticities with input expenditure shares to construct firm-level markups.

7.1 Estimation procedure

We replicate the empirical design in Raval (2023).¹⁰ Firms’ log real revenue satisfies a Cobb–Douglas revenue production function in capital, labor, and materials, with productivity evolving as a linear AR(1). As in Raval (2023), we implement the ACF control-function estimator using a third-order polynomial approximation in (k, ℓ, m) for the proxy/state relationship. We then re-estimate the same specification with our one-step CUE estimator and report identification-robust inference alongside conventional large-sample approximations.

We follow Raval (2023) in variable construction, sample selection, and the ACF implementation. Output y_{it} is log real revenue and inputs $(k_{it}, \ell_{it}, m_{it})$ are log real capital, labor, and materials. In ENIA, materials are measured as intermediate consumption. In Compustat, materials are not observed and are proxied by COGS–XLR, where COGS (cost of goods sold) bundles materials, direct labor, and overhead, while XLR measures labor and related expenses with incomplete coverage. We estimate the model separately by country and industry; within each sample, the maintained specification and instrument set are held fixed across estimators, so that differences in estimates reflect the choice of estimator (ACF versus CUE) rather than changes in specification.

¹⁰Data construction details and industry definitions follow App. A of Raval (2023).

7.2 Specification testing

Alongside identification-robust inference we assess specification. As discussed in Section 5.4.1, wide robust confidence sets do not by themselves indicate misspecification: they are consistent with a correctly specified model in which the proxy provides limited information about productivity. To separate this from genuine violations of the moment conditions, we report both the conventional Hansen J -statistic and the Kleibergen KJ -statistic (25) evaluated at the CUE estimate.

The two statistics differ precisely in their behavior under weak identification: the Hansen J is $\chi^2_{df-d_\theta}$ only when identification is strong, whereas KJ retains this limiting distribution regardless of identification strength. A finding of non-rejection for KJ alongside wide AR-GMM confidence sets provides evidence that the overidentifying restrictions are satisfied and that the identification difficulty stems from limited proxy signal rather than from model misspecification. Rejection of KJ , by contrast, points to an underlying violation of the moment conditions—possible sources include factor-biased productivity (which would cause the output elasticities to vary with ω_{it}), within-industry heterogeneity in the production function, or errors in variable construction that contaminate the moment conditions.

7.3 Markups

We follow the production approach to markup estimation rooted in the wedge logic of Hall (1988) and implemented at the firm level by De Loecker and Warzynski (2012b) and subsequent work (e.g. De Loecker et al., 2020). Let $v \in \{\ell, m\}$ be a flexible input and let θ_{it}^v denote its output elasticity implied by the estimated production function. Under static cost minimization in v conditional on state inputs, no adjustment costs, and firms taking input prices as given, the first-order condition implies

$$\mu_{it} = \frac{\theta_{it}^v}{s_{it}^v}, \quad s_{it}^v = \frac{p_{it}^v v_{it}}{P_{it} Y_{it}},$$

where s_{it}^v is the revenue share of input expenditure and μ_{it} is the price markup over marginal cost. When both labor and materials satisfy the maintained conditions, μ_{it} is invariant to the choice of v .

Raval (2023) uses this invariance as a specification check and documents systematic discrepancies between labor- and materials-based markups across datasets, emphasizing the sensitivity of production-approach markups to technology non-neutrality and measurement. De Ridder et al. (2022) further emphasize that with financial-statement data—where prices and quantities are typ-

ically unobserved— revenue-based production function estimates need not identify the level of markups, although they can remain informative about dispersion and changes. Motivated by these insights, we report markups based on both labor and materials shares and relate their behavior to the strength of identification in the underlying elasticity estimates.

A key feature of our setting is that, in the presence of weak identification, the elasticities (and hence markups) are naturally viewed as partially identified objects. We therefore complement the standard “plug-in” markup series based on point estimates with uncertainty summaries that propagate identification-robust elasticity bounds into markups, both for time-series moments and for the cross-sectional distribution.

Let $(\underline{\beta}^v, \overline{\beta}^v)$ denote the identification-robust confidence set for the elasticity of flexible input v (obtained from the Subset- S procedure). Holding input shares fixed, this induces observation-level markup bounds

$$\underline{\mu}_{it}^v = \frac{\underline{\beta}^v}{s_{it}^v}, \quad \overline{\mu}_{it}^v = \frac{\overline{\beta}^v}{s_{it}^v},$$

where we enforce ordering of the bounds in finite samples. For each year we summarize these bounds by their cross-sectional means, $\mathbb{E}_t[\underline{\mu}_{it}^v]$ and $\mathbb{E}_t[\overline{\mu}_{it}^v]$, and overlay the corresponding mean based on the CUE point estimates, $\mathbb{E}_t[\widehat{\mu}_{it}^v]$. We report changes relative to an initial base year to focus on time variation.

To characterize uncertainty in the full cross-sectional distribution of markups, we treat each firm-year markup as interval-valued, $\mu_{it}^v \in [\underline{\mu}_{it}^v, \overline{\mu}_{it}^v]$, and construct two complementary distributional summaries.

We summarize these interval-valued markups with the *exact interval-mixture density* implied by the observation-level bounds. Specifically, each firm-year contributes a uniform density on its interval $[\underline{\mu}_{it}^v, \overline{\mu}_{it}^v]$, and the cross-sectional density is the equal-weight mixture of these observation-specific uniforms. This yields the density

$$f_{\mu^v}(x) = \frac{1}{N_t} \sum_{i=1}^{N_t} \frac{\mathbf{1}\{x \in [\underline{\mu}_{it}^v, \overline{\mu}_{it}^v]\}}{\overline{\mu}_{it}^v - \underline{\mu}_{it}^v},$$

defined wherever the interval width is positive. We overlay this exact interval-mixture density with the kernel density of the point-estimate markups $\widehat{\mu}_{it}^v$ computed from CUE elasticities. The comparison illustrates how much of the cross-sectional distribution is driven by identification uncertainty in the elasticities, versus dispersion in the plug-in point estimates themselves.

7.4 Data

We use two datasets that differ in measurement quality. The Chilean data are from ENIA and are plant-level, survey-based observations for *Manufacture of food products* (ISIC 311) over 1979–1996, comprising approximately 16,000 plant–year observations. Materials are recorded as intermediate consumption and are cleanly separated from labor compensation, yielding a relatively precise proxy for productivity.

The U.S. data are from Compustat and are firm-level accounting data for *Food Manufacturing* (NAICS 31) over 1970–2005, comprising approximately 1,000 firm–year observations. Relative to ENIA, key inputs are measured with more error—most importantly, materials are constructed from accounting aggregates—so the proxy–state mapping is weaker and weak-identification concerns are correspondingly more salient in the U.S. sample.

7.5 Chile (ENIA)

Table C.5 reports results for the Chilean ENIA sample (ISIC 311). The estimates are stable across estimators. Under ACF, elasticities are $\hat{\beta}_k = 0.040$, $\hat{\beta}_\ell = 0.123$, and $\hat{\beta}_m = 0.905$, implying mildly increasing returns to scale (RTS= 1.068). The CUE estimates are similar— $\hat{\beta}_k = 0.014$, $\hat{\beta}_\ell = 0.111$, and $\hat{\beta}_m = 0.947$ —with RTS= 1.072.

The Cragg–Donald minimum-eigenvalue statistic is $T_{CD} = 336.0$ (χ^2_2 critical value at 95%: 5.99, $p < 0.001$), decisively rejecting the null of underidentification and confirming that the Jacobian has full rank. This is reflected in the confidence sets: the Subset- S intervals for β_ℓ and β_m are only modestly wider than the corresponding Wald intervals, and the J–K intervals are of comparable width. Specifically, the J–K 95% confidence sets are [0.010, 0.032] for β_k , [0.058, 0.118] for β_ℓ , [0.936, 0.976] for β_m , and [1.030, 1.090] for RTS. These are somewhat tighter than the corresponding Subset- K intervals, as the J pretest prunes the spurious flat region of the K statistic at values where the moment conditions do not hold. The Hansen J -test does not reject at CUE ($J = 0.166$, $\text{df} = 1$, $p = 0.684$), consistent with correct specification.

Bootstrap diagnostics reinforce this benign picture. The Shapiro–Wilk test fails to reject normality for all three elasticities ($p = 0.506$ for β_k , $p = 0.369$ for β_ℓ , and $p = 0.926$ for β_m), the corresponding Jarque–Bera tests also fail to reject ($p = 0.940$, 0.648, and 0.987), and the Doornik–Hansen joint test likewise does not reject ($p = 0.823$). Taken together, these diagnostics are consistent with approximately standard large-sample behavior in the Chilean sample: the Jacobian is far

from rank-deficient, the robust confidence sets are well-bounded, and the bootstrap draws do not exhibit the pronounced non-Gaussian features seen in the U.S. data.

Figure D.3 plots the evolution of average markups in Chile (ENIA, ISIC 311), separately for labor- and materials-based markups. The solid line reports the annual mean of the plug-in markups constructed from the CUE point estimates. The shaded band reports the corresponding year-by-year bounds obtained by propagating the identification-robust elasticity intervals into observation-level markup intervals and averaging within year.¹¹ The bounds are relatively tight throughout the sample and track the point series closely, indicating that the time variation in average markups is informative and not driven by identification uncertainty in the underlying elasticities.

Figure D.4 compares two estimates of the cross-sectional distribution of markups in a representative year (1996). The solid curve reports the exact interval-mixture density obtained by averaging the observation-specific uniform densities on $[\underline{\mu}_{it}^v, \bar{\mu}_{it}^v]$, while the dashed curve reports the kernel density of plug-in CUE markups $\hat{\mu}_{it}^v$. The two densities are close, and the interval-mixture distribution is only slightly more dispersed, suggesting that identification uncertainty has limited distributional consequences in this sample. Relative to the U.S. case, the markup inference is well-behaved, consistent with a strong proxy signal in ENIA where materials are measured directly as intermediate consumption.

7.6 U.S. (Compustat)

Table C.6 reports results for the U.S. Compustat sample (NAICS 31). The estimates are considerably less stable than in the Chilean survey data. Under ACF, elasticities are $\hat{\beta}_k = 0.329$, $\hat{\beta}_\ell = 0.132$, and $\hat{\beta}_m = 0.561$, implying approximately constant returns to scale (RTS= 1.022). In contrast, the CUE estimates differ sharply: $\hat{\beta}_k = 0.728$, $\hat{\beta}_\ell = 1.507$, and $\hat{\beta}_m = -1.000$, implying mildly increasing returns to scale (RTS= 1.235). This large divergence between ACF and CUE—with sign reversal for materials and a tenfold increase in the labor elasticity—is characteristic of weak identification: when the GMM criterion is nearly flat, small changes in weighting produce large swings in the minimizer.

Identification-robust inference directly confirms this picture. The Cragg–Donald minimum-eigenvalue statistic fails to reject underidentification ($T_{CD} = 0.218$, $\text{df} = 2$, $p = 0.897$), indicating that the Jacobian is near-rank-deficient and the moment conditions contain very limited information about the structural parameters. The Subset- S confidence sets are correspondingly wide:

¹¹In each year we compute $\underline{\mu}_{it}^v = \underline{\beta}^v / s_{it}^v$ and $\bar{\mu}_{it}^v = \bar{\beta}^v / s_{it}^v$ and then take cross-sectional means. The figure reports changes relative to the first sample year.

$[-0.470, 1.730]$ for β_k , $[-7.290, 3.910]$ for β_ℓ , $(-\infty, -0.200]$ for β_m , and $[-2.360, 2.040]$ for RTS. The Wald intervals substantially understate this sampling uncertainty. Bootstrap normality is strongly rejected for all elasticities (Shapiro–Wilk $p < 0.001$ for each; Doornik–Hansen joint $p \approx 0$), consistent with the highly non-Gaussian sampling behavior predicted by Proposition 8 under weak identification.

The Subset- JK confidence sets are wide and left-unbounded for all parameters: $(-\infty, 1.730]$ for β_k , $(-\infty, 3.910]$ for β_ℓ , $(-\infty, 6.800]$ for β_m , and $(-\infty, 7.240]$ for RTS. These sets are broader than Subset- S on the right, which is expected when the J pretest accepts values where the K statistic is spuriously small but the moment conditions do not strongly reject. In this weakly identified setting the Subset- JK intervals should be read as confirming that the data provide virtually no lower bound on the elasticities, rather than as a failure of the test: the upper bounds are informative while the left boundary is driven by the grid.

On specification, the conventional Hansen J -test does not reject at the CUE estimate ($J = 3.44$, $\text{df} = 1$, $p = 0.064$). Because identification is weak, the Hansen J need not have the correct size, and non-rejection alone is not conclusive. The main empirical issue is therefore not model misspecification but a weak proxy signal: with Compustat materials constructed as $\text{COGS} - \text{XLR}$, the proxy link to latent productivity is noisy, and the resulting moment conditions have limited leverage for pinning down the elasticities.

Figure D.5 reports the U.S. (Compustat, NAICS 31) time series of mean markups, constructed analogously to the Chilean case. The solid line is the annual mean of plug-in CUE markups $\mathbb{E}_t[\hat{\mu}_{it}^v]$, while the shaded band plots the corresponding mean bounds $\mathbb{E}_t[\underline{\mu}_{it}^v]$ and $\mathbb{E}_t[\overline{\mu}_{it}^v]$ induced by the identification-robust elasticity intervals. In stark contrast to ENIA, the bounds are extremely wide throughout the sample and often extend far into economically implausible regions, so that changes in average markups are only weakly informative once elasticity uncertainty is propagated. This pattern is consistent with the weak-identification features already evident in the elasticity estimates and their robust confidence sets.

Figure D.6 illustrates the same issue for the cross-sectional distribution in 1996. The dashed kernel density based on plug-in CUE markups is tightly concentrated, reflecting the cross-sectional dispersion in the point estimates. By contrast, the solid exact interval-mixture density based on interval markups $\mu_{it}^v \in [\underline{\mu}_{it}^v, \overline{\mu}_{it}^v]$ is extremely diffuse and nearly flat over a wide range. This gap indicates that, in the U.S. sample, a substantial part of what appears as a precise distribution under

plug-in markups is not supported once identification-robust elasticity uncertainty is accounted for: the data deliver limited information about the level and dispersion of markups in this setting, in line with concerns raised by De Ridder et al. (2022) for financial-statement data.

8 Conclusion

This paper revisits control-function estimation of production functions through the lens of weak identification. Formal point identification can hold even when the proxy for productivity is noisy—as we show following Doraszelski and Li (2025), the standard moment conditions retain their identifying power under a suitable index-sufficiency condition. In practice, however, formal identification and the practical strength of identification are distinct. The information content of the moment conditions can be low in empirically relevant settings: the proxy’s informativeness about latent productivity can be limited by optimization frictions, heterogeneous input prices, supply and demand shocks, or the imprecision inherent in constructing proxy variables from accounting aggregates. When this is the case, the GMM criterion becomes nearly flat in economically meaningful directions, leading to unstable estimates and non-Gaussian sampling behavior, so conventional Wald-type inference can be unreliable.

To address this, we cast the control-function estimator in a GMM framework that makes the role of the moment Jacobian explicit and clarifies how limited proxy signal translates into near-rank failure. Building on this characterization, we implement three complementary tools for applied work: (i) a bootstrap normality diagnostic that screens for weak-signal environments, (ii) identification-robust confidence sets based on AR/S-type statistics (and associated Subset- S confidence sets) computed from a continuously updated GMM objective with cluster-robust variance, and (iii) the Kleibergen KJ -statistic, which provides a valid test of the overidentifying restrictions regardless of identification strength and thus allows us to distinguish limited proxy informativeness from genuine model misspecification. These procedures remain valid regardless of identification strength and provide a practical alternative to reporting only conventional standard errors and Wald intervals.

Empirically, the contrast between Chilean survey data (ENIA) and U.S. financial-statement data (Compustat) illustrates the stakes. In ENIA, where materials are measured directly as intermediate consumption, elasticity estimates are stable across estimators, bootstrap diagnostics are consistent with approximately standard behavior, and the specification tests do not reject—consistent with

a correctly specified model with a strong proxy signal. Propagating identification-robust elasticity intervals into markups yields relatively tight bounds on both the time series of mean markups and the cross-sectional distribution. In Compustat, where materials must be constructed from accounting aggregates (e.g. COGS – XLR), estimates are markedly less stable and identification-robust intervals become very wide. The Kleibergen KJ -test does not reject, supporting the interpretation that the difficulty is one of weak identification rather than misspecification: the moment conditions appear valid, but the proxy provides limited leverage for pinning down the elasticities. Once identification uncertainty is propagated into markups, both levels and distributional features become only weakly informative relative to plug-in series computed from point estimates.

For applied researchers using OP/LP/ACF-style designs—especially in settings where proxies are constructed from aggregated data—our recommendation is: report a weak-signal diagnostic together with identification-robust confidence sets, accompany these with the KJ specification test to assess whether the moment conditions are credible, and when markups or other nonlinear functionals are of interest, explicitly propagate identification-robust elasticity uncertainty into those objects. This helps separate genuine economic dispersion and time variation from artifacts of weak identification and nonstandard sampling.

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Appendices

A Proofs of Section 4

Lemma 10. Suppose $m_n(\vartheta_w, \vartheta_{s0})$ is continuously differentiable in ϑ_w on a compact convex set Θ_w , $m_n(\vartheta_{w0}, \vartheta_{s0}) = 0$, and $\partial m_n(\vartheta_w, \vartheta_{s0}) / \partial \vartheta'_w = M_{w,n}(\vartheta_w, \vartheta_{s0})$. If $M_{w,n}(\vartheta_w, \vartheta_{s0}) = n^{-1/2}C(\vartheta_w, \vartheta_{s0}) + o(n^{-1/2})$ uniformly for $\vartheta_w \in \Theta_w$ and for a bounded function $C(\vartheta_w, \vartheta_{s0})$ uniformly for $\vartheta_w \in \Theta_w$, then

$$m_n(\vartheta_w, \vartheta_{s0}) = n^{-1/2}m_1(\vartheta_w) + o(n^{-1/2}) \quad \text{for all } \vartheta_w \in \Theta_w,$$

where

$$m_1(\vartheta_w) := \int_0^1 C(\vartheta_{w0} + t(\vartheta_w - \vartheta_{w0}), \vartheta_{s0})(\vartheta_w - \vartheta_{w0}) dt.$$

In particular, $m_n(\vartheta_w, \vartheta_{s0}) = n^{-1/2}m_1(\vartheta_w) + o(n^{-1/2})$ uniformly on Θ_w .

Proof. Fix $\vartheta_w \in \Theta_w$ and define the line segment $\vartheta_{w,t} := \vartheta_{w0} + t(\vartheta_w - \vartheta_{w0}) \in \Theta_w$ for $t \in [0, 1]$. By the fundamental theorem of calculus and the assumed interchange of differentiation and expectation,

$$\begin{aligned} m_n(\vartheta_w, \vartheta_{s0}) - m_n(\vartheta_{w0}, \vartheta_{s0}) &= \int_0^1 \frac{d}{dt} m_n(\vartheta_{w,t}, \vartheta_{s0}) dt \\ &= \int_0^1 \frac{\partial m_n(\vartheta_{w,t}, \vartheta_{s0})}{\partial \vartheta'_w} (\vartheta_w - \vartheta_{w0}) dt \\ &= \int_0^1 M_{w,n}(\vartheta_{w,t}, \vartheta_{s0})(\vartheta_w - \vartheta_{w0}) dt. \end{aligned}$$

Using $m_n(\vartheta_{w0}, \vartheta_{s0}) = 0$ and $M_{w,n}(\vartheta_{w,t}, \vartheta_{s0}) = n^{-1/2}C(\vartheta_{w,t}, \vartheta_{s0}) + o(n^{-1/2})$ yields

$$m_n(\vartheta_w, \vartheta_{s0}) = n^{-1/2} \int_0^1 C(\vartheta_{w0} + t(\vartheta_w - \vartheta_{w0}), \vartheta_{s0})(\vartheta_w - \vartheta_{w0}) dt + o(n^{-1/2}) = n^{-1/2}m_1(\vartheta_w) + o(n^{-1/2}),$$

which proves the claim. $\square$

Proof of Proposition 7. We have that,

$$\begin{aligned} m_n(\vartheta_w, \vartheta_s) &= m_n(\vartheta_w, \vartheta_{s0}) + R(\vartheta_w)(\vartheta_s - \vartheta_{s0}) \\ &\quad + \underbrace{\left(M_{w,n}(\vartheta_w, \vartheta_s) - M_{w,n}(\vartheta_w, \vartheta_{s0}) \right)}_{O(n^{-1/2})} (\vartheta_w - \vartheta_{w0}) + r_n(\vartheta_w, \vartheta_s) - r_n(\vartheta_w, \vartheta_{s0}), \end{aligned}$$

The $O(n^{-1/2})$ -rate comes from Assumption 4 i), compactness of Θ_w , Assumption 4, (iv): the original remainder r_n is uniformly $o(n^{-1/2})$. By Lemma 10, we can write

$$m_n(\vartheta_w, \vartheta_s) = n^{-1/2}m_1(\vartheta_w) + R(\vartheta_w)(\vartheta_s - \vartheta_{s0}) + O(n^{-1/2})$$

Consistency. By a uniform LLN and $W_n \xrightarrow{p} W_0$,

$$\frac{1}{n} Q_n(\vartheta_w, \vartheta_s) = Q_{0,n}(\vartheta_w, \vartheta_s) + o_p(1)$$

where, $Q_{0,n}(\vartheta_w, \vartheta_s) = m_n(\vartheta_w, \vartheta_s)' W_0 m_n(\vartheta_w, \vartheta_s)$. By Lemma 10, the decomposition above, and

boundedness of $R(\vartheta_w)$ on Θ_w ,

$$Q_{0,n} = (\vartheta_s - \vartheta_{s0})' R(\vartheta_w)' W_0 R(\vartheta_w) (\vartheta_s - \vartheta_{s0}) + O(n^{-1/2}),$$

uniformly on $\Theta_w \times \Theta_s$.

Since W_0 is positive definite and $R(\vartheta_w)$ has full column rank uniformly on Θ_w , there exists $c > 0$ such that

$$(\vartheta_s - \vartheta_{s0})' R(\vartheta_w)' W_0 R(\vartheta_w) (\vartheta_s - \vartheta_{s0}) \geq c \|\vartheta_s - \vartheta_{s0}\|^2 \quad \forall \vartheta_w \in \Theta_w.$$

Fix any $\epsilon > 0$. Then for all $\vartheta_w \in \Theta_w$ and all ϑ_s such that $\|\vartheta_s - \vartheta_{s0}\| \geq \epsilon$,

$$Q_{0,n}(\vartheta_w, \vartheta_s) \geq c\epsilon^2 + O(n^{-1/2}).$$

Hence, for n sufficiently large,

$$\inf_{\vartheta_w \in \Theta_w, \|\vartheta_s - \vartheta_{s0}\| \geq \epsilon} Q_{0,n}(\vartheta_w, \vartheta_s) \geq \frac{c\epsilon^2}{2}.$$

Note that $Q_{0,n}(\vartheta_{w0}, \vartheta_{s0}) = 0$, so the population criterion is uniformly separated away from zero outside any ϵ -neighborhood of ϑ_{s0} . And since $\frac{1}{n}Q_n$ is uniformly approximated by $Q_{0,n}$, standard extremum-estimator arguments imply $\hat{\vartheta}_s \xrightarrow{P} \vartheta_{s0}$.

$\sqrt{n}$ -consistency. By the previous step, $\hat{\vartheta}_s \xrightarrow{P} \vartheta_{s0}$, and by Assumption 6, $\lambda_{\min}(W_n) \xrightarrow{P} \lambda_{\min}(W_0) > 0$. Moreover, under Assumption 5, $\sup_{\vartheta \in \Theta_w \times \Theta_s} \|\psi_n(\vartheta)\| = O_p(1)$.

Since $\hat{\vartheta}$ minimizes $Q_n(\vartheta)$, we have $Q_n(\hat{\vartheta}) \leq Q_n(\vartheta_0)$, hence

$$\lambda_{\min}(W_n) \|\psi_n(\hat{\vartheta}) + \sqrt{n} m_n(\hat{\vartheta})\|^2 \leq Q_n(\hat{\vartheta}) \leq Q_n(\vartheta_0) = \psi_n(\vartheta_0)' W_n \psi_n(\vartheta_0) = O_p(1).$$

Therefore, $\|\psi_n(\hat{\vartheta}) + \sqrt{n} m_n(\hat{\vartheta})\| = O_p(1)$, and hence $\|\sqrt{n} m_n(\hat{\vartheta})\| = O_p(1)$.

Using the same expansion as above of $m_n(\vartheta_w, \vartheta_s)$ in ϑ_s ,

$$\sqrt{n} m_n(\hat{\vartheta}_w, \hat{\vartheta}_s) = \sqrt{n} m_n(\hat{\vartheta}_w, \vartheta_{s0}) + R(\hat{\vartheta}_w) \sqrt{n}(\hat{\vartheta}_s - \vartheta_{s0}) + O(1).$$

By Lemma 10, $\sup_{\vartheta_w \in \Theta_w} \|\sqrt{n} m_n(\vartheta_w, \vartheta_{s0})\| = O(1)$, so $\|\sqrt{n} m_n(\hat{\vartheta}_w, \vartheta_{s0})\| = O(1)$. Combining with $\|\sqrt{n} m_n(\hat{\vartheta})\| = O_p(1)$ gives

$$\|R(\hat{\vartheta}_w) \sqrt{n}(\hat{\vartheta}_s - \vartheta_{s0})\| = O_p(1).$$

Since $[C(\vartheta_w, \vartheta_s) R(\vartheta_w)]$ has full column rank $d_w + d_s$ uniformly on $\Theta_w \times \Theta_s$ by Assumption 4(iii), $R(\vartheta_w)$ itself has full column rank d_s uniformly on Θ_w (if $R(\vartheta_w)x = 0$ then $[C R] \begin{pmatrix} 0 \\ x \end{pmatrix} = 0$, contradicting full column rank). Since R is continuous on the compact set Θ_w , there exists $c_R > 0$ such that $\|R(\vartheta_w)x\| \geq c_R \|x\|$ for all $x \in \mathbb{R}^{d_s}$ and all $\vartheta_w \in \Theta_w$. Applying this with $\vartheta_w = \hat{\vartheta}_w$ yields $\sqrt{n}(\hat{\vartheta}_s - \vartheta_{s0}) = O_p(1)$. $\square$

Proof of Proposition 8. Let $\tilde{\vartheta}_s = \vartheta_{s0} + \lambda/\sqrt{n}$ for fixed $\lambda \in \mathbb{R}^{d_s}$. Using Lemma 10 we can write

$$\sqrt{n} m_n(\vartheta_w, \tilde{\vartheta}_s) = m_1(\vartheta_w) + R(\vartheta_w)\lambda + O(1),$$

uniformly in $\vartheta_w \in \Theta_w$. Moreover, since Θ_w is compact and $m_1(\vartheta_w)$ and $R(\vartheta_w)$ are bounded on Θ_w ,

$$\sup_{\vartheta_w \in \Theta_w} \|\sqrt{n} m_n(\vartheta_w, \tilde{\vartheta}_s)\| = O(1).$$

By Assumption 5, $\psi_n \Rightarrow \psi_0$, where ψ_0 has almost surely bounded sample paths. Since $\tilde{\vartheta}_s - \vartheta_{s0} \rightarrow$

0, asymptotic stochastic equicontinuity of ψ_n implies

$$\sup_{\vartheta_w \in \Theta_w} \|\psi_n(\vartheta_w, \tilde{\vartheta}_s) - \psi_n(\vartheta_w, \vartheta_{s0})\| \xrightarrow{p} 0.$$

Hence

$$\sup_{\vartheta_w \in \Theta_w} \|\psi_n(\vartheta_w, \tilde{\vartheta}_s)\| = O_p(1).$$

Therefore, using

$$\sqrt{n} \bar{\phi}_n(\vartheta_w, \tilde{\vartheta}_s) = \psi_n(\vartheta_w, \tilde{\vartheta}_s) + \sqrt{n} m_n(\vartheta_w, \tilde{\vartheta}_s),$$

we obtain

$$\sup_{\vartheta_w \in \Theta_w} \|\sqrt{n} \bar{\phi}_n(\vartheta_w, \tilde{\vartheta}_s)\| = O_p(1).$$

Since $W_n \xrightarrow{p} W_0$, we have $\|W_n\| = O_p(1)$, and therefore

$$\begin{aligned} \sup_{\vartheta_w \in \Theta_w} Q_n(\vartheta_w, \tilde{\vartheta}_s) &= \sup_{\vartheta_w \in \Theta_w} \left[\left(\sqrt{n} \bar{\phi}_n(\vartheta_w, \tilde{\vartheta}_s) \right)' W_n \left(\sqrt{n} \bar{\phi}_n(\vartheta_w, \tilde{\vartheta}_s) \right) \right] \\ &\leq \|W_n\| \sup_{\vartheta_w \in \Theta_w} \|\sqrt{n} \bar{\phi}_n(\vartheta_w, \tilde{\vartheta}_s)\|^2 = O_p(1). \end{aligned}$$

□

B Monte Carlo Design

We simulate $n = 1000$ firms observed for $T = 10$ consecutive periods. Unless otherwise noted, parameter values are taken from ACF. The parameters are calibrated so that roughly 95% of the cross-sectional variation in capital is between firms and a fixed-effects regression of k_{it} on l_{it} delivers $R^2 \approx 0.50$.

B.1 Choice of Labour and Material Inputs

Conditional on its information set, the firm solves

$$\max_{L_{it}, M_{it}} \alpha K_{it}^{\beta_k} L_{it}^{\beta_l} \exp(\omega_{it}) \quad \text{s.t.} \quad M_{it} \geq \beta_m^{-1} \alpha K_{it}^{\beta_k} L_{it}^{\beta_l} \exp(\omega_{it}),$$

yielding in optimality

$$\begin{aligned} L_{it}^* &= \left(\frac{\alpha \beta_l}{W_{it}} \right)^{\frac{1}{1-\beta_l}} K_{it}^{\frac{\beta_k}{1-\beta_l}} \exp\left(\frac{\omega_{it}}{1-\beta_l} \right) \\ M_{it}^* &= \beta_m^{-1} \alpha K_{it}^{\beta_k} L_{it}^{*\beta_l} \exp(\omega_{it}) \end{aligned}$$

Following ACF, we contaminate L_{it}^* with iid optimisation noise to break the functional dependence issue.

$$L_{it} = L_{it}^* \exp(\xi_{it}^l)$$

where ξ_{it}^l follows a normal distribution with mean zero and a standard deviation of 0.1. ACF chooses a standard deviation of the optimisation error of 0.37. We deliberately adopt a smaller value. A larger variance indeed strengthens identification by injecting more cross-sectional variation into the labour input and thereby mitigating—but not eliminating—the weak-instrument problem. For expositional

clarity, however, a modest reduction in the variance still breaks the functional dependence at the heart of the ACF critique while keeping the simulated distributions readily interpretable.

B.2 Investment Choice and Capital Accumulation

Capital is a dynamic input. Investment is chosen at $t-1$ and the capital stock evolves as

$$K_{i,t+1} = (1 - \delta)K_{it} + I_{it}, \quad c_i(I_{it}) = \frac{\phi_i}{2} I_{it}^2,$$

where $\phi_i > 0$ is firm-specific; we draw ϕ_i^{-1} lognormally with cross-sectional standard deviation 0.6. This implies a strong heterogeneity, with a substantial mass of firms having very high adjustment costs. Firms maximize the expected present value of profits net of adjustment costs. With the static labor choice contaminated by multiplicative optimization error,

$$L_{it} = L_{it}^* \exp(\xi_{it}^l), \quad \xi_{it}^l \sim \mathcal{N}(0, \sigma_{\xi_l}^2) \text{ i.i.d.},$$

the (expected) marginal value of one more unit of installed capital next period is

$$\text{MVPK}_{i,t+1} = \left(\frac{\beta_k}{1 - \beta_l} \right) \alpha^{\frac{1}{1-\beta_l}} W_{i,t+1}^{-\frac{\beta_l}{1-\beta_l}} \exp\left(\frac{\omega_{i,t+1}}{1 - \beta_l} \right) \left[\beta_l^{\frac{\beta_l}{1-\beta_l}} e^{\frac{1}{2}\beta_l^2 \sigma_{\xi_l}^2} - \beta_l^{\frac{1}{1-\beta_l}} e^{\frac{1}{2}\sigma_{\xi_l}^2} \right].$$

The term in square brackets is due to the optimization-error in labor. The Euler equation with quadratic adjustment costs yields a closed-form policy:

$$I_{it} = \frac{\beta}{\phi_i} \sum_{\tau=0}^{\infty} (\beta(1 - \delta))^{\tau} \mathbb{E}_t[\text{MVPK}_{i,t+1+\tau}].$$

We assume $\ln W_{it}$ follows an independent covariance-stationary AR(1):

$$\ln W_{t+h} = \rho_W^h \ln W_t + \zeta_{t+h},$$

with $\zeta_{t+h} \sim \mathcal{N}(0, \sigma_{\zeta}^2 \sum_{j=0}^{h-1} \rho_W^{2j})$, independent of current information. Then conditional log-normality implies, for $h = \tau + 1$,

$$\mathbb{E}_t \left[e^{\frac{\omega_{t+h}}{1-\beta_l}} W_{t+h}^{-\frac{\beta_l}{1-\beta_l}} \right] = \exp \left(\frac{\rho^h}{1 - \beta_l} \omega_t - \frac{\rho_W^h \beta_l}{1 - \beta_l} \ln W_t + \frac{1}{2} \frac{V_{\omega, h-1}}{(1 - \beta_l)^2} + \frac{1}{2} \frac{\beta_l^2 V_{W, h-1}}{(1 - \beta_l)^2} \right),$$

where the forecast-error variances have closed forms (for $|\rho|, |\rho_W| < 1$):

$$V_{\omega, h-1} = \sigma_{\xi}^2 \sum_{j=0}^{h-1} \rho^{2j} = \sigma_{\xi}^2 \frac{1 - \rho^{2h}}{1 - \rho^2}, \quad V_{W, h-1} = \sigma_{\zeta}^2 \sum_{j=0}^{h-1} \rho_W^{2j} = \sigma_{\zeta}^2 \frac{1 - \rho_W^{2h}}{1 - \rho_W^2}.$$

Substituting the last display into the Euler sum gives a fully explicit expression for I_{it} .

$$I_{it} = \frac{\beta}{\phi_i} \frac{\beta_k}{1 - \beta_{\ell}} \alpha^{\frac{1}{1-\beta_{\ell}}} B_{\xi} \sum_{\tau=0}^{\infty} [\beta(1 - \delta)]^{\tau} \exp \left(\frac{\rho^{\tau+1}}{1 - \beta_{\ell}} \omega_{it} - \frac{\rho_W^{\tau+1} \beta_{\ell}}{1 - \beta_{\ell}} \ln W_{it} + \frac{1}{2} V_{\tau} \right),$$

with

$$B_{\xi} = \beta_{\ell}^{\frac{\beta_{\ell}}{1-\beta_{\ell}}} e^{\frac{1}{2}\beta_{\ell}^2 \sigma_{\xi_{\ell}}^2} - \beta_{\ell}^{\frac{1}{1-\beta_{\ell}}} e^{\frac{1}{2}\sigma_{\xi_{\ell}}^2}, \quad V_{\tau} = \frac{V_{\omega, \tau}}{(1 - \beta_{\ell})^2} + \frac{\beta_{\ell}^2 V_{W, \tau}}{(1 - \beta_{\ell})^2}$$

B.3 Steady-State Initialisation

All firms start at $K_{i0} = 0$. We simulate the model forward until the cross-section of capital stocks converges to the stationary distribution implied by the policy rule; 10 consecutive periods from that steady state are retained for estimation.

B.4 Estimation

We apply the ACF and the LP estimators described in ACF Appendix A. The CUE is described in Section 5. We use moment vector

$$f_{it}(\theta) = z_{it} (y_{it} - \alpha - \beta_k k_{it} - \beta_l l_{it} - \rho(y_{it-1} - \alpha - \beta_k k_{it-1} - \beta_l l_{it-1})), \quad (\text{B.1})$$

with instrument vector $z_{it} = (l_{it-1}, k_{it}, m_{it-1})$.

C Tables

Table C.1: Monte-Carlo estimates ($\sigma_{\xi_l} = 0.37$)

Meas.	ACF				LP				CUE			
	β_l		β_k		β_l		β_k		β_l		β_k	
	Coef.	S.D.	Coef.	S.D.	Coef.	S.D.	Coef.	S.D.	Coef.	S.D.	Coef.	S.D.
0.0	0.599	0.010	0.401	0.016	0.600	0.003	0.391	0.015	0.600	0.017	0.400	0.006
0.1	0.604	0.010	0.409	0.015	0.678	0.003	0.318	0.012	0.599	0.019	0.400	0.006
0.2	0.609	0.011	0.409	0.016	0.725	0.003	0.270	0.012	0.601	0.021	0.400	0.006
0.5	0.621	0.014	0.405	0.017	0.797	0.003	0.187	0.016	0.601	0.026	0.400	0.007

1000 replications. True values of β_l and β_k are 0.6 and 0.4, respectively.

Table C.2: Rejection frequencies at 5 % nominal level ($\sigma_{\xi_l} = 0.37$)

Meas.	Wald	S	KLM	DH	CD reject
0.0	0.058	0.052	—	0.089	—
0.1	0.054	0.056	—	0.079	—
0.2	0.054	0.057	—	0.099	—
0.5	0.062	0.051	—	0.127	—

1000 replications. Wald: joint $\chi^2(3)$ at CUE estimate. S : joint Anderson–Rubin $\chi^2(3)$ evaluated at true θ_0 . KLM: Kleibergen–Moreira–Lindner joint $\chi^2(3)$ at true θ_0 . DH: Doornik–Hansen multivariate normality test on bootstrap distribution. CD reject: rejection frequency of the Cragg–Donald rank test ($T_{CD} \sim \chi^2(1)$); rejection indicates identification.

Table C.3: Monte-Carlo estimates ($\sigma_{\xi_l} = 0.1$)

Meas.	ACF				LP				CUE			
	β_l		β_k		β_l		β_k		β_l		β_k	
	Coef.	S.D.	Coef.	S.D.	Coef.	S.D.	Coef.	S.D.	Coef.	S.D.	Coef.	S.D.
0.000	0.595	0.038	0.405	0.041	0.600	0.010	0.391	0.017	0.599	0.023	0.400	0.007
0.100	0.638	0.068	0.365	0.070	0.907	0.005	2.54e4	2.42e5	0.598	0.031	0.400	0.008
0.200	0.675	0.084	0.329	0.087	0.945	0.004	1.28e6	1.27e6	0.591	0.043	0.402	0.010
0.500	-7.40e8	1.09e10	7.39e8	1.09e10	0.972	0.003	4.41e5	4.00e5	0.581	0.076	0.404	0.015

1000 replications. True values of β_l and β_k are 0.6 and 0.4, respectively. Values in red indicate explosive behavior.

Table C.4: Rejection frequencies at 5 % nominal level ($\sigma_{\xi_i} = 0.10$)

Meas.	Wald	S	KLM	DH	CD reject
0.0	0.064	0.060	0.060	0.065	1.000
0.1	0.059	0.048	0.048	0.079	1.000
0.2	0.075	0.062	0.062	0.177	1.000
0.5	0.128	0.050	0.050	0.427	1.000

See notes to Table C.2.

Table C.5: Elasticity estimates: Chile ENIA, ISIC 311

Parameter	Estimates		95% CI			BS normality
	ACF	CUE	Wald	Subset- S	Subset- JK	SW p -value
β_k	0.040	0.014	[0.006, 0.022]	[0.004, 0.026]	[0.010, 0.032]	0.506
β_ℓ	0.123	0.111	[0.087, 0.134]	[0.082, 0.142]	[0.058, 0.118]	0.369
β_m	0.905	0.947	[0.930, 0.964]	[0.924, 0.968]	[0.936, 0.976]	0.926
Returns to scale	1.068	1.072	[1.060, 1.085]	[1.056, 1.090]	[1.030, 1.090]	—

Cragg–Donald minimum-eigenvalue statistic: $T_{CD} = 336.0$ (df= 2, $p < 0.001$). Hansen J -test at CUE: $J = 0.166$ (df= 1, $p = 0.684$).

Notes: Wald confidence intervals are based on the cluster-robust variance at the CUE estimate. “Subset- S ” reports identification-robust confidence sets obtained by inverting the S-statistic. “Subset- JK ” reports the combined Kleibergen J–K confidence sets ($\alpha_K = 0.04$, $\alpha_J = 0.01$), which correct for the spurious power decline of the K statistic by pretesting moment validity. “SW p -value” reports the bootstrap p -value of a Shapiro–Wilk normality test for the corresponding standardized bootstrap distribution.

Table C.6: Elasticity estimates: U.S. Compustat, NAICS 31

Parameter	Estimates		95% CI			BS normality
	ACF	CUE	Wald	Subset- S	Subset- JK	SW p -value
β_k	0.329	0.728	[−0.796, 2.253]	[−0.470, 1.730]	[<−10, 1.730]	< 0.001
β_ℓ	0.132	1.507	[−0.362, 3.376]	[−7.290, 3.910]	[<−10, 3.910]	< 0.001
β_m	0.561	−1.000	[−2.406, 0.406]	[<−10, −0.200]	[<−10, 6.800]	< 0.001
Returns to scale	1.022	1.235	[0.652, 1.819]	[−2.360, 2.040]	[<−10, 7.240]	—

Cragg–Donald minimum-eigenvalue statistic: $T_{CD} = 0.218$ (df= 2, $p = 0.897$); fails to reject underidentification. Hansen J -test at CUE: $J = 3.44$ (df= 1, $p = 0.064$).

Notes: Wald confidence intervals are based on the cluster-robust variance at the CUE estimate. “Subset- S ” reports identification-robust confidence sets obtained by inverting the S-statistic. “Subset- JK ” reports the combined Kleibergen J–K confidence sets ($\alpha_K = 0.04$, $\alpha_J = 0.01$). In both columns, a leading “<” followed by a value (e.g. [<−10, 1.730]) indicates the set extends to the left boundary of the estimation grid; the reported number is the grid boundary used. “SW p -value” reports the bootstrap p -value of a Shapiro–Wilk normality test for the corresponding standardized bootstrap distribution.

D Figures

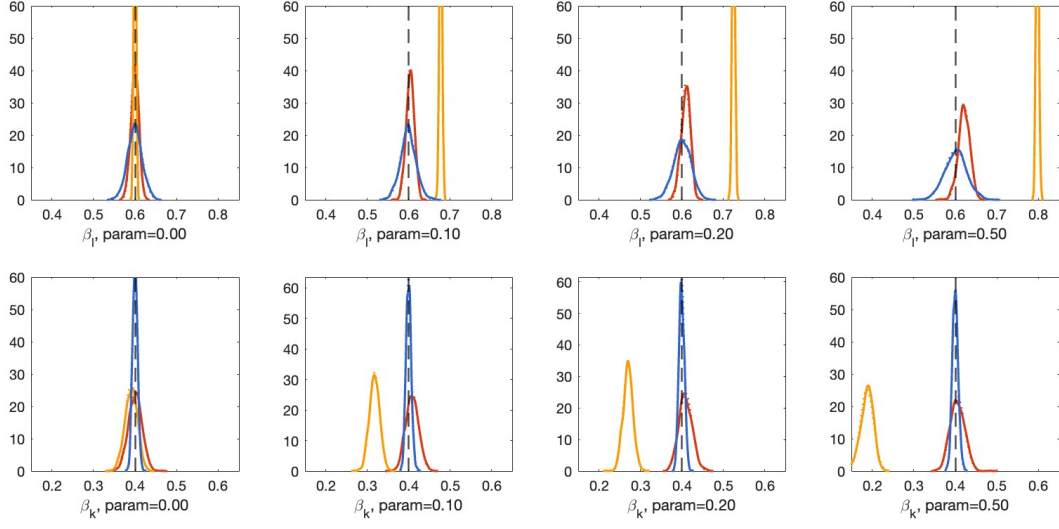


Figure D.1: 1000 replications. $\sigma_{\varepsilon_l} = 0.37$. Empirical densities of the estimates for β_l (top row) and β_k (bottom row) across measurement-error levels 0%, 10%, 20%, 50% (left to right). Methods: ACF (red), LP (orange), CUE (blue). The dashed vertical line marks the true parameter value. Empirical densities: solid. Normal pdf: dotted. 1000 replications.

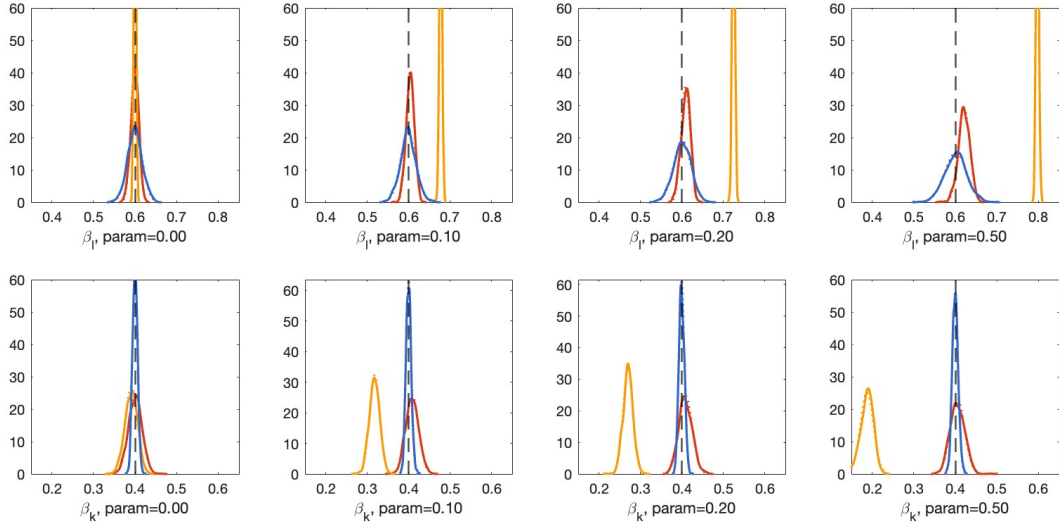


Figure D.2: 1000 replications. $\sigma_{\xi_l} = 0.10$. Empirical densities of the estimates for β_l (top row) and β_k (bottom row) across measurement-error levels 0%, 10%, 20%, 50% (left to right). Methods: ACF (red), LP (orange), CUE (blue). The dashed vertical line marks the true parameter value. Empirical densities: solid. Normal pdf: dotted. 1000 replications.

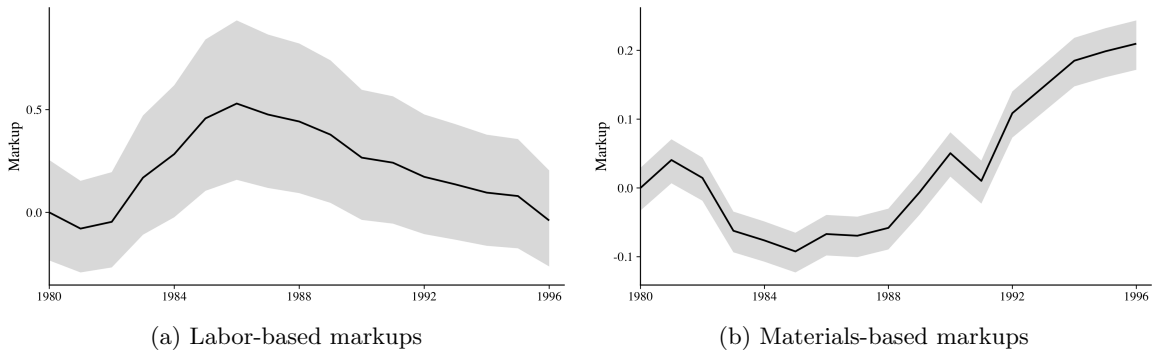


Figure D.3: Chile (ENIA, ISIC 311): changes in mean markups with identification-robust bounds. Solid line: mean plug-in CUE markups. Shaded band: mean bounds induced by identification-robust elasticity intervals. Series are plotted relative to the first sample year.

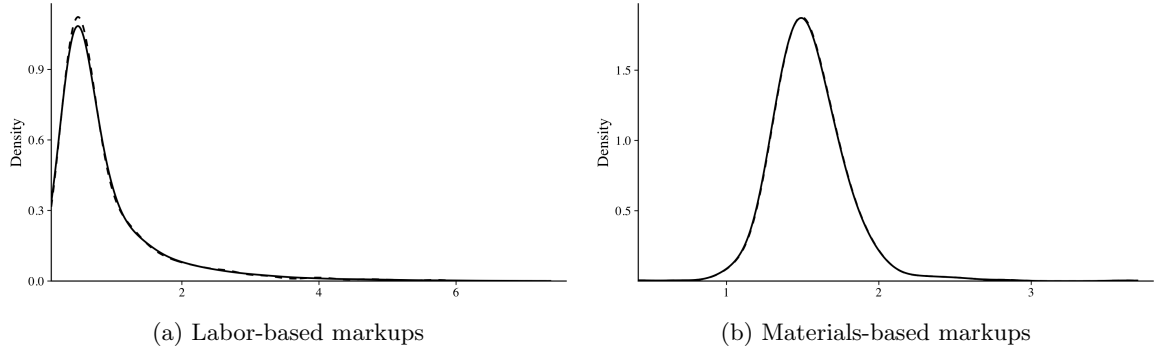


Figure D.4: Chile (ENIA, ISIC 311): markup distributions in 1996. Dashed/red line: imputation-mixture kernel density from draws $\mu_{it}^{v,*} \sim \text{Unif}[\underline{\mu}_{it}^v, \bar{\mu}_{it}^v]$ pooled across repetitions. Solid/blue line: kernel density of plug-in CUE markups $\hat{\mu}_{it}^v$.

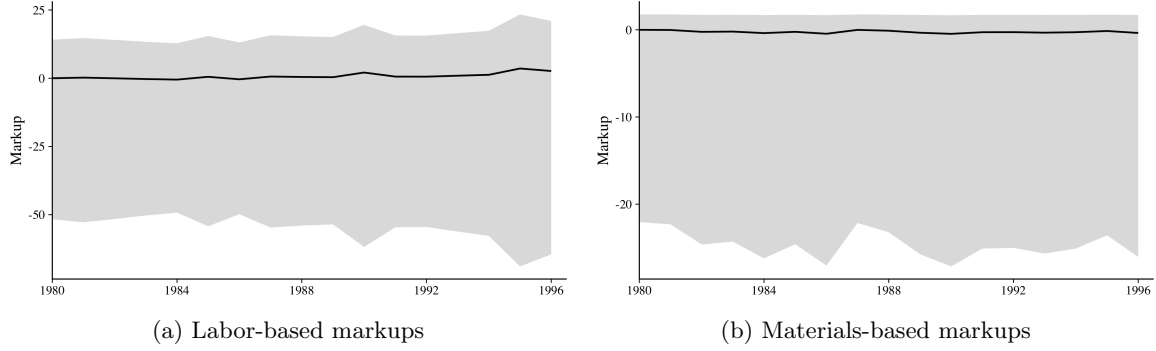


Figure D.5: U.S. (Compustat, NAICS 31): changes in mean markups with identification-robust bounds. Solid line: mean plug-in CUE markups. Shaded band: mean bounds induced by identification-robust elasticity intervals. Series are plotted relative to the first sample year.

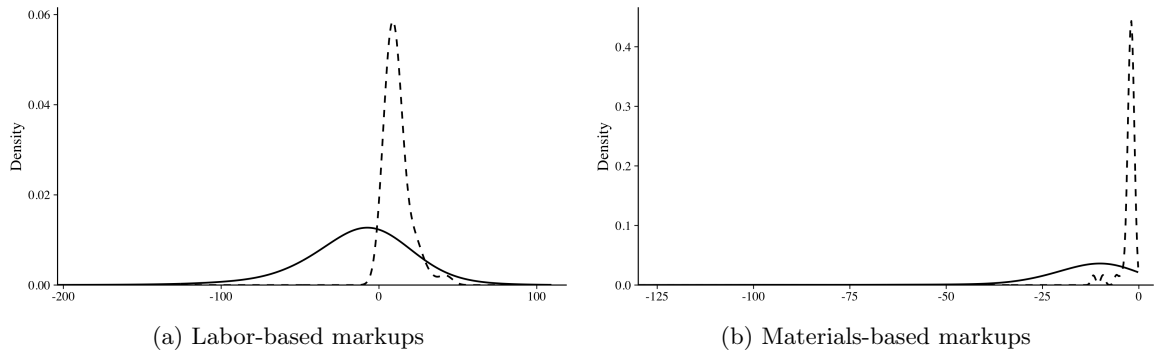


Figure D.6: U.S. (Compustat, NAICS 31): markup distributions in 1996. Dashed line: imputation-mixture kernel density from draws $\mu_{it}^{v,*} \sim \text{Unif}[\underline{\mu}_{it}^v, \bar{\mu}_{it}^v]$ pooled across repetitions. Solid line: kernel density of plug-in CUE markups $\hat{\mu}_{it}^v$.